厦门大学信息学院 本科选修课
2024-2025 第二学期

模 式 识 别
Pattern Recognition

主 讲：王程教授、陈龙彪副教授
厦门大学信息学院计算机科学与技术系
人工智能引论101计划系列课程



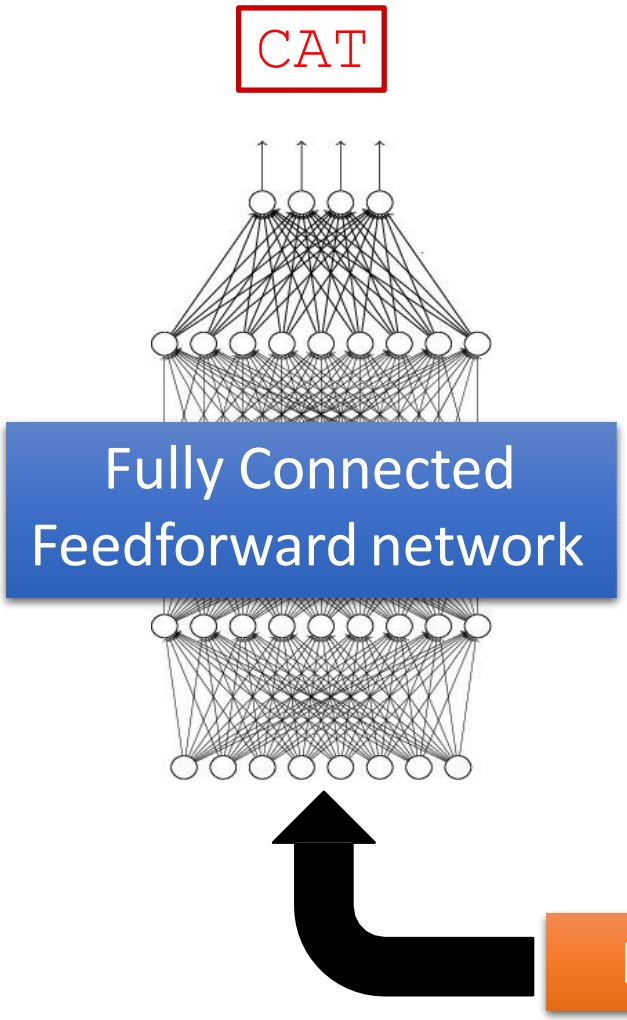
神经网络

- 深度神经网络
- 卷积神经网络
- 循环神经网络
- …… ……

卷积神经网络（CNN）

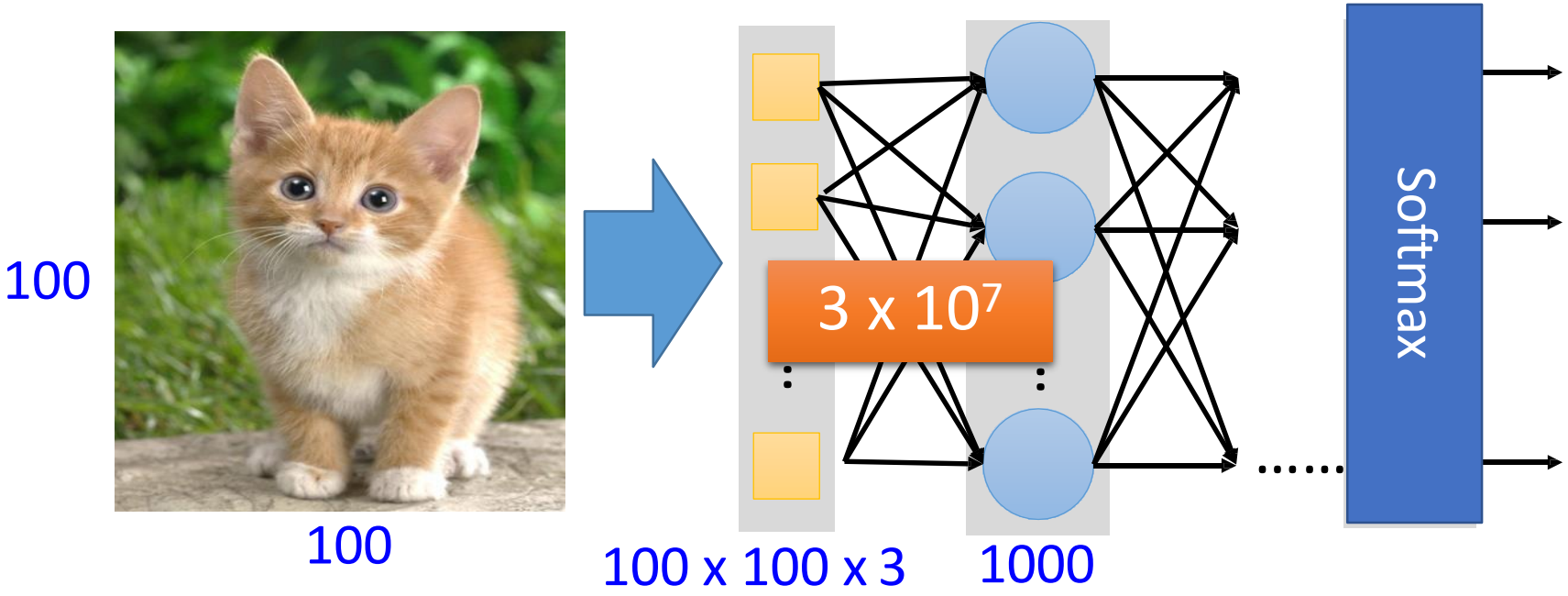
CNN网络结构




CAT
Convolution
Max Pooling
Convolution
Max Pooling
Flatten
Fully Connected
Feedforward network
可以重复
多次

为什么CNN 可以用于图像识别?

- 处理图像时，全连接网络的输入层会非常大



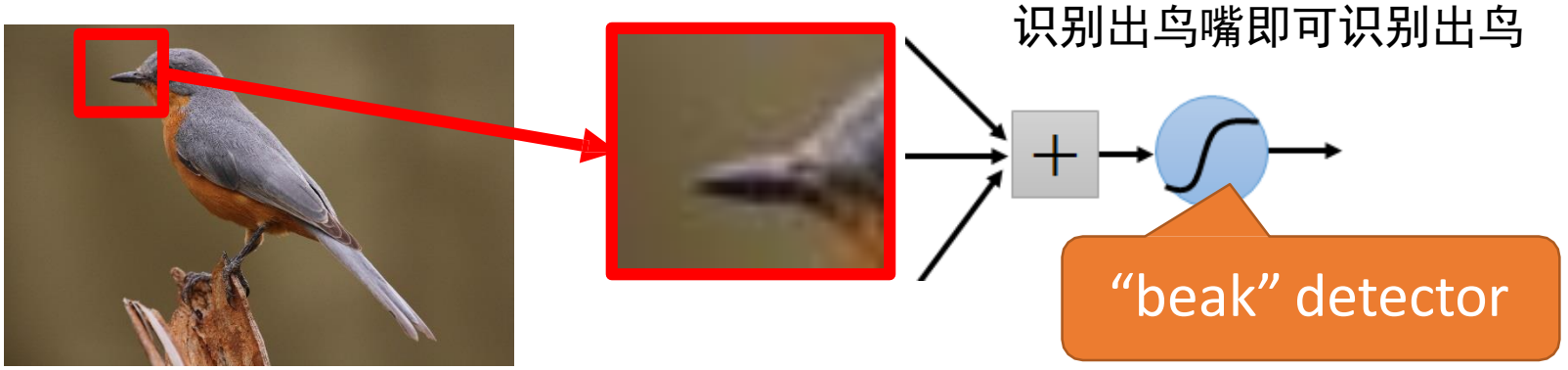
可以通过考虑图像识别的性质来简化全连接网络吗?

为什么CNN 用于图像识别?

1. 一些模式比整张图片小得多

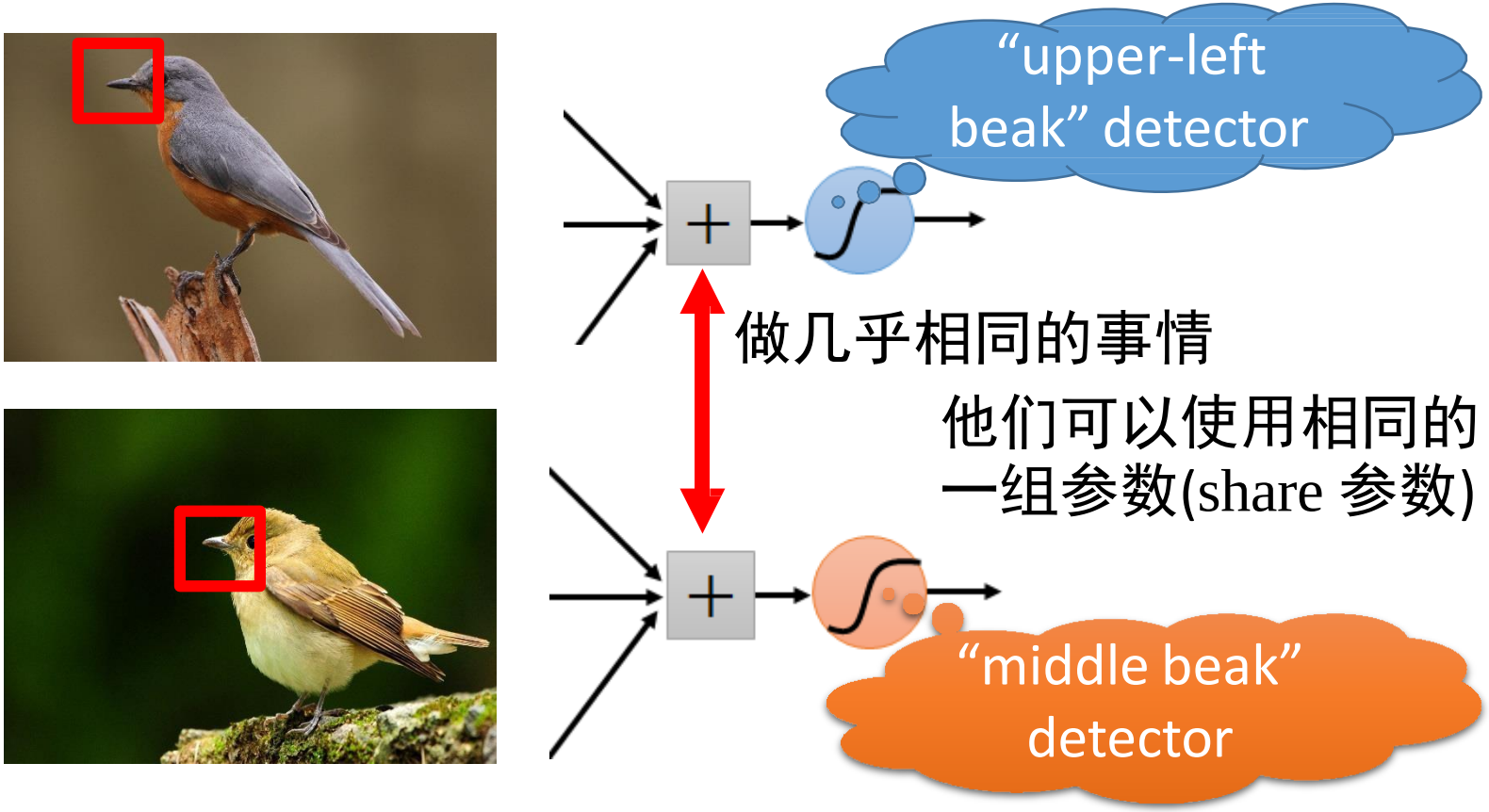
一个神经元不需要看到整个图像去发现模式

通过较少的参数连接到小区域



为什么CNN 用于图像识别?

2. 同样的模式可能出现在图像的不同区域



为什么CNN 用于图像识别?

3. 对图像二次采样不会改变对物体的认知

- 只要不是太过分（电子包浆）



二次采样



我们可以对图像进行二次采样以使图像变小

→ 更少的参数来处理图像

CNN网络结构



属性 1

- 一些模式比整张图片小得多

属性 2

- 同样的模式可能出现在图像的不同区域

属性 3

- 对图像的二次采样不会改变图像中的物体

Convolution

Max Pooling

Convolution

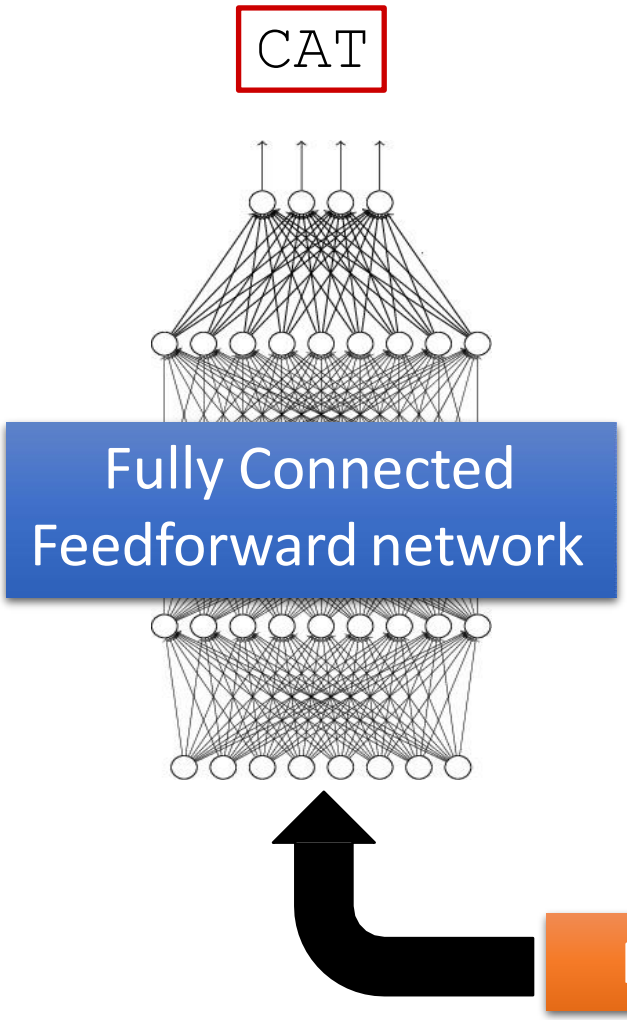
Max Pooling

可以重复多次

Flatten

CNN网络结构




CAT
Fully Connected
Feedforward network
Convolution
Max Pooling
Convolution
Max Pooling
Flatten
可以重复
多次

CNN – Convolution

- Kernel: 卷积核

注意：这些kernel都是要学习的参数！

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Kernel 1 矩阵

-1	1	-1
-1	1	-1
-1	1	-1

Kernel 2 矩阵

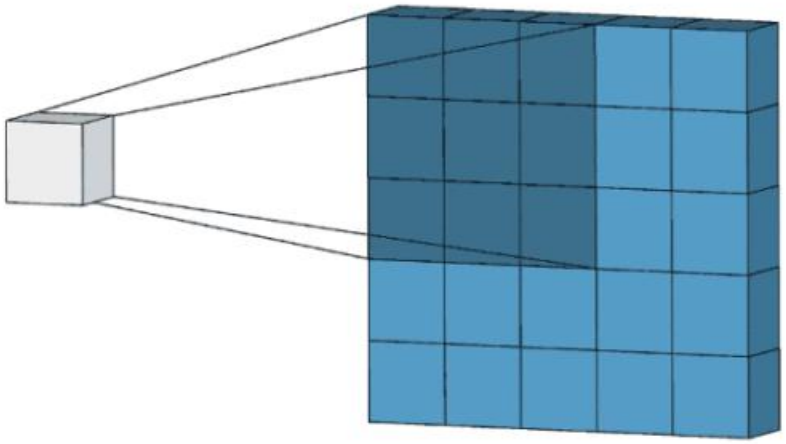
⋮

每个 kernel 探测一个小的模式(3 x 3)

属性一：一些模式比整张图片小得多

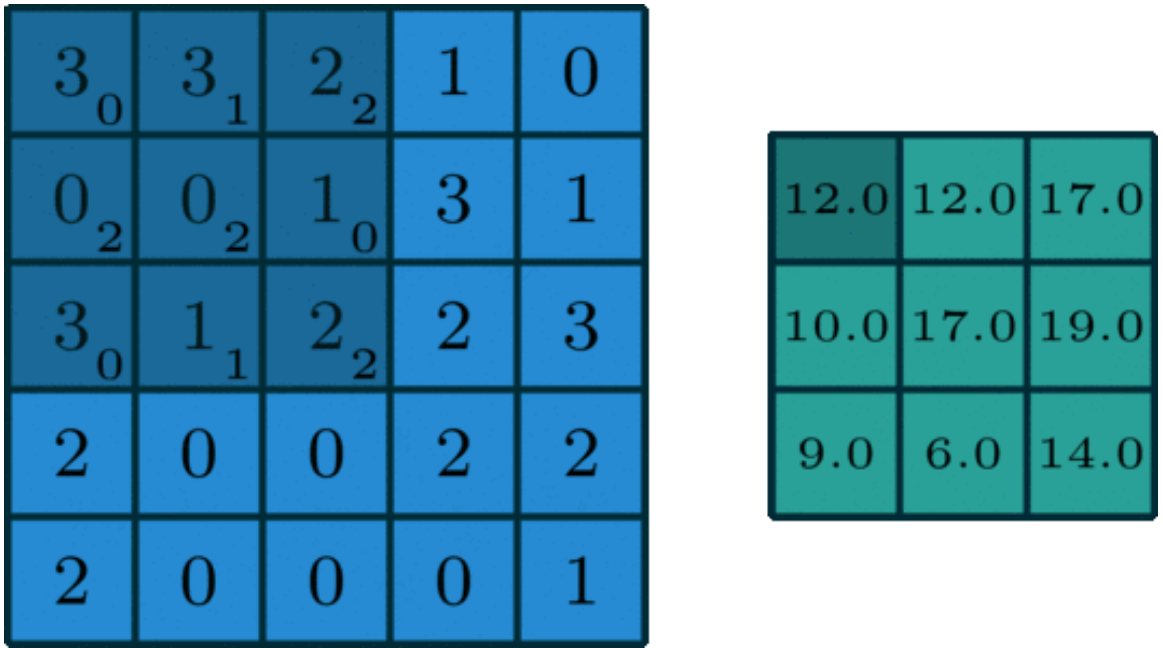
CNN – Convolution

- **卷积是一个相当简单的操作：**先从一个小小的权重矩阵，也就是卷积核（kernel）开始，让它逐步在二维输入数据上“扫描”。卷积核“滑动”的同时，计算权重矩阵和扫描所得的数据矩阵的乘积，然后把结果汇总成一个输出像素



CNN – Convolution

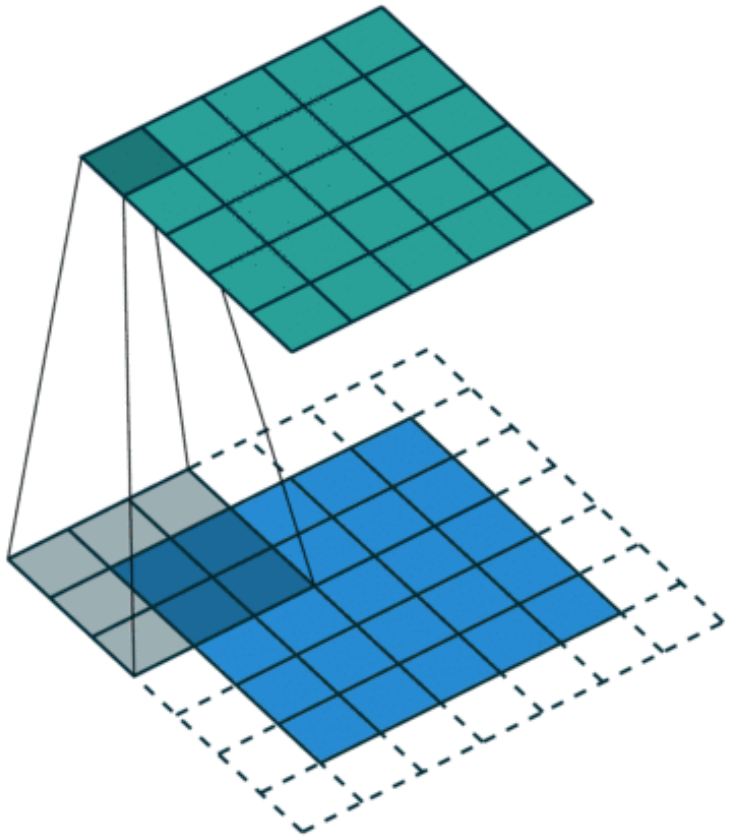
- 输出的特征是原输入特征的加权和
- 权重是卷积核自带的值
- 像素位置所处的地方大致相同



CNN – Convolution

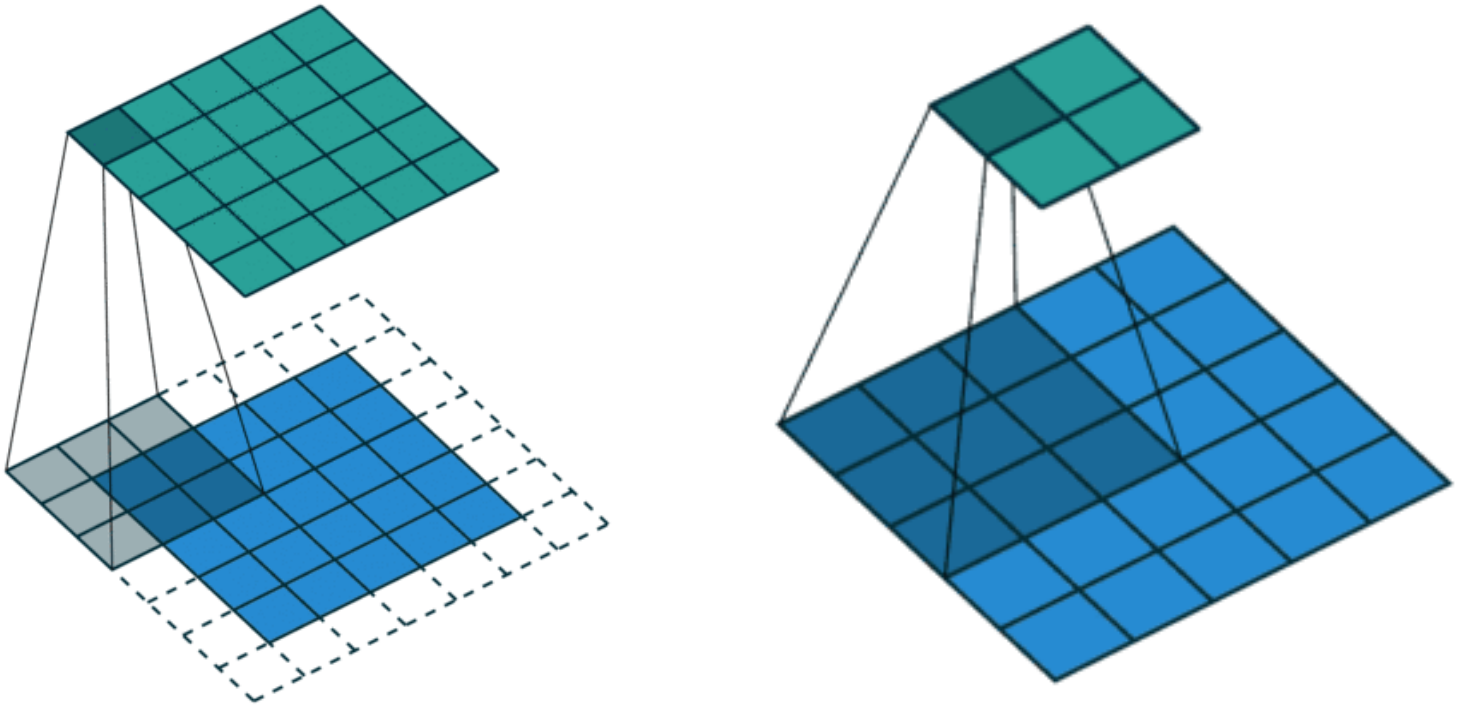
Padding: 填充

- 用额外的“假”像素填充边缘（值一般为0）



CNN – Convolution

- **Stride: 步长**
  - 每次滑动的行数和列数称为Stride



CNN – Convolution

步长=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Kernel

-3	-1	-3	-1
-3	-1	0	-3
-3	-3	0	1
-3	-2	-2	-1

属性二：同样的模式可能出现在图像的不同区域

CNN – Convolution

对于每一个 kernel 重复卷积操作

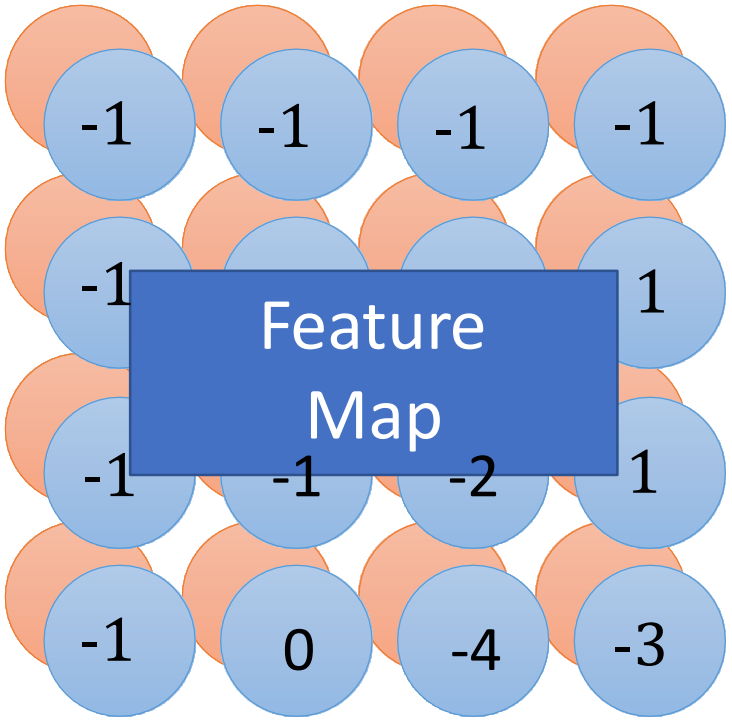
-1	1	-1
-1	1	-1
-1	1	-1

Kernel 2

步长=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

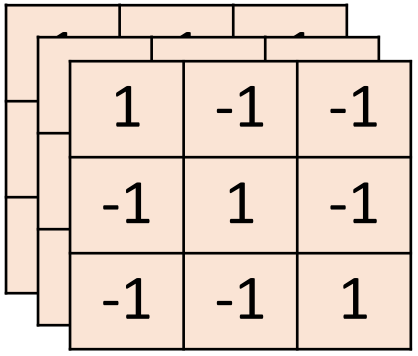
6 x 6 image



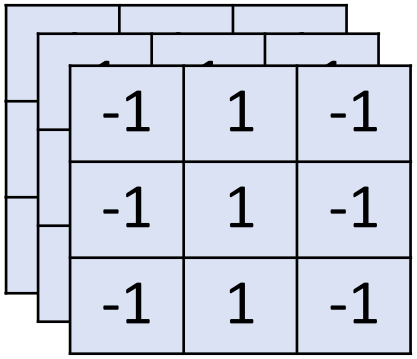
4 x 4 image * 2

CNN – Convolution

彩色图像

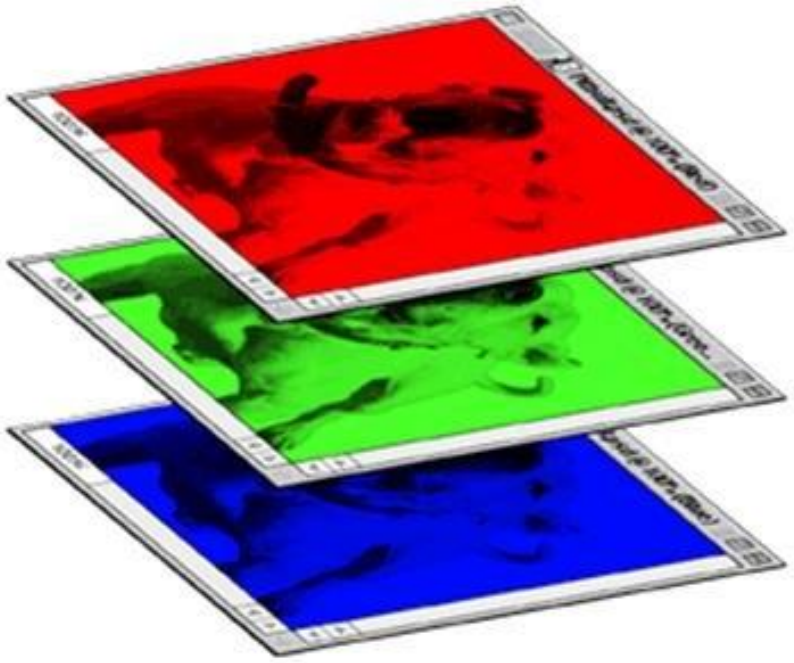


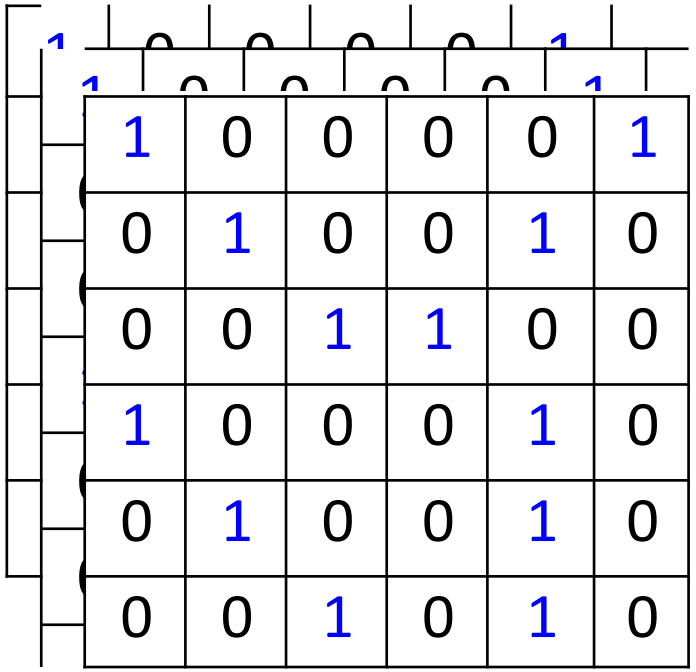
Kernel 1



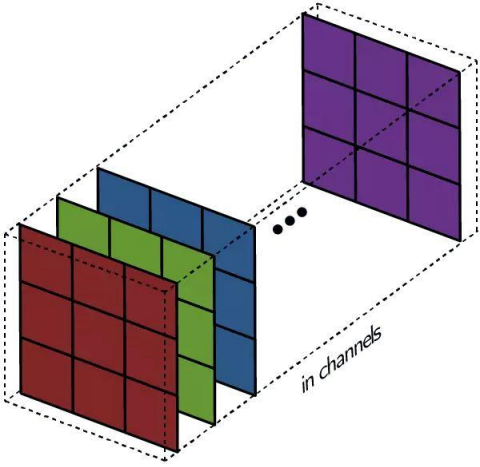
Kernel 2

三通道





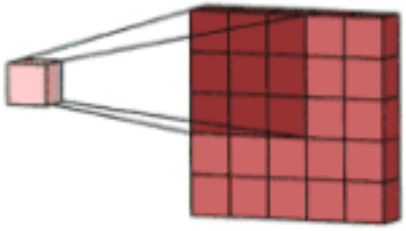
CNN – Convolution

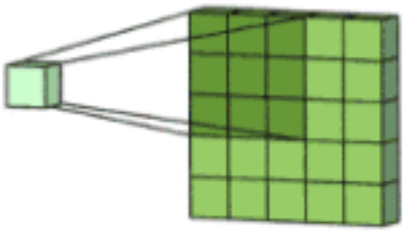


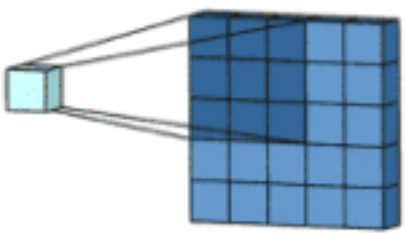
- Channel: 通道



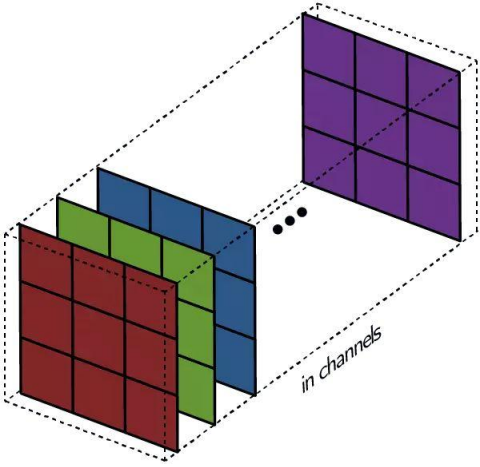








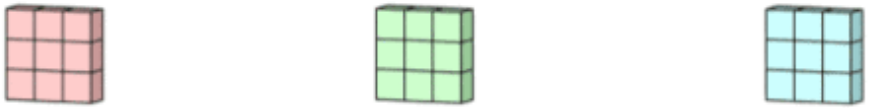
CNN – 彩色图像



- Channel: 通道





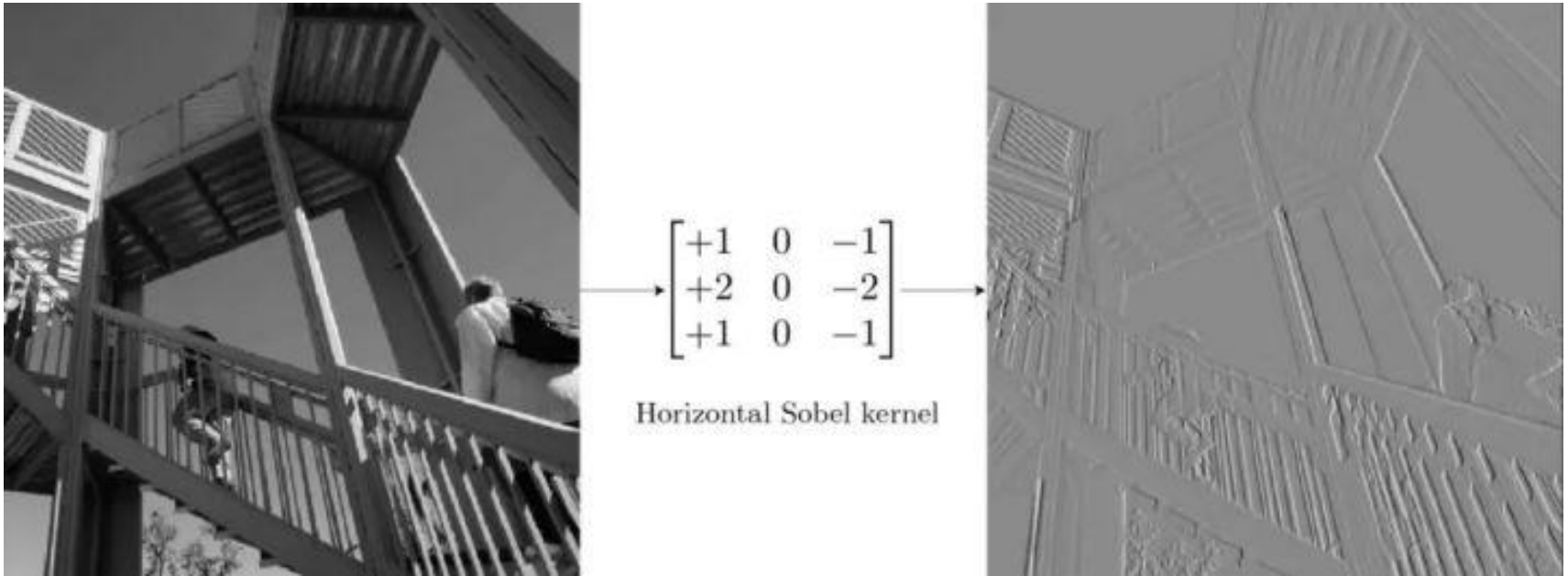


CNN – Convolution

卷积核的例子

边缘检测核：Sobel edge detection

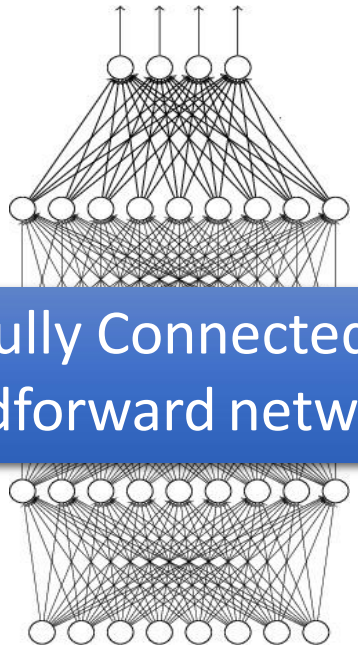
索伯算子最早是由美国计算机科学家艾尔文·索伯（Irwin Sobel）及盖瑞·费德曼（Gary Feldman）于1968年在斯坦福大学的人工智能实验室(SAIL)所提出，因此为了表扬他们的贡献，才用他们的名字命名。



The whole CNN



cat dog ……



Fully Connected Feedforward network

Convolution

Max Pooling

Convolution

Max Pooling

Can repeat many times

Flatten

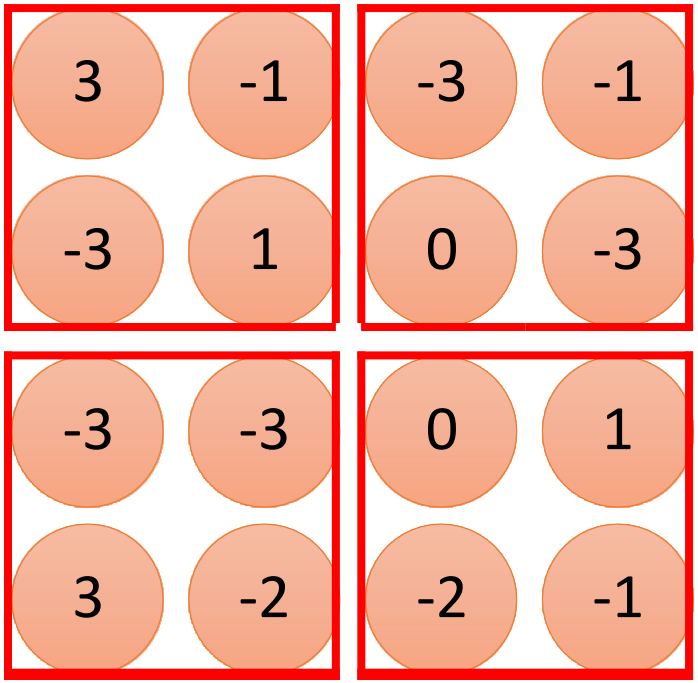
CNN – Max Pooling

1	-1	-1
-1	1	-1
-1	-1	1

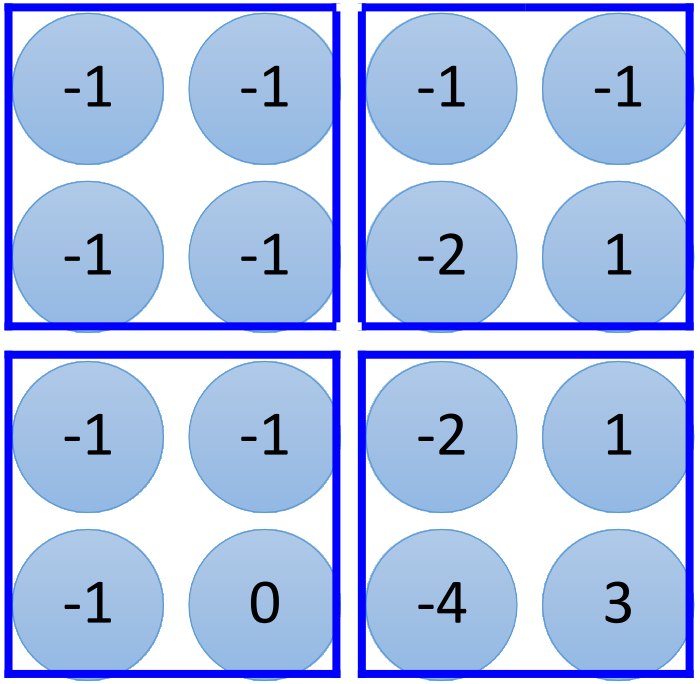
Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

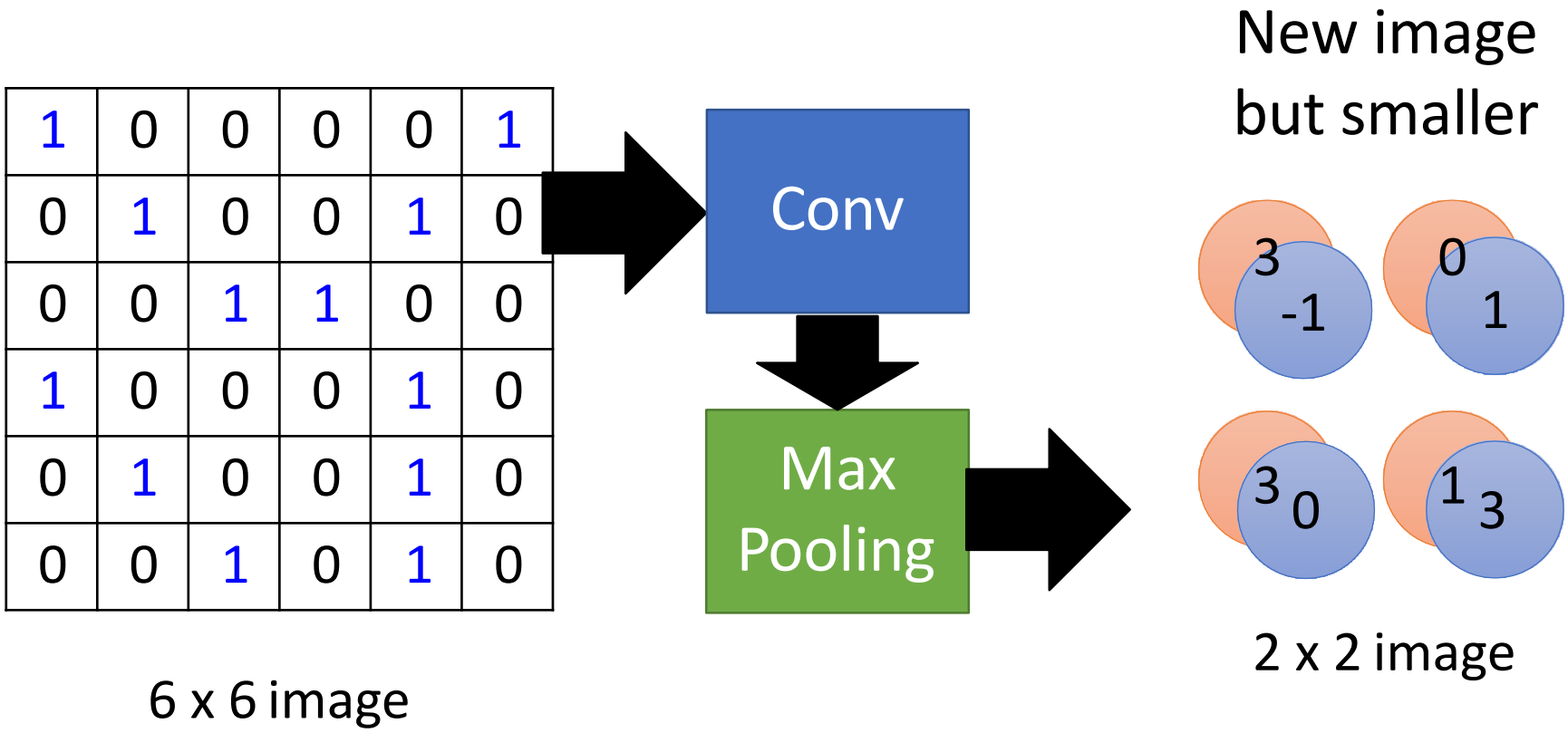


4 x 4 feature map



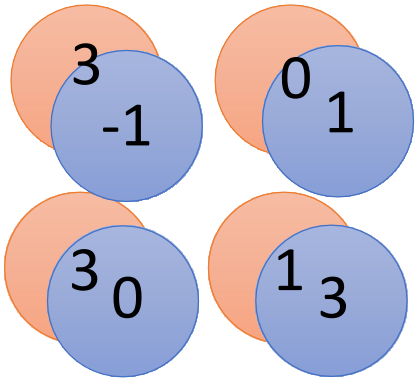
4 x 4 feature map

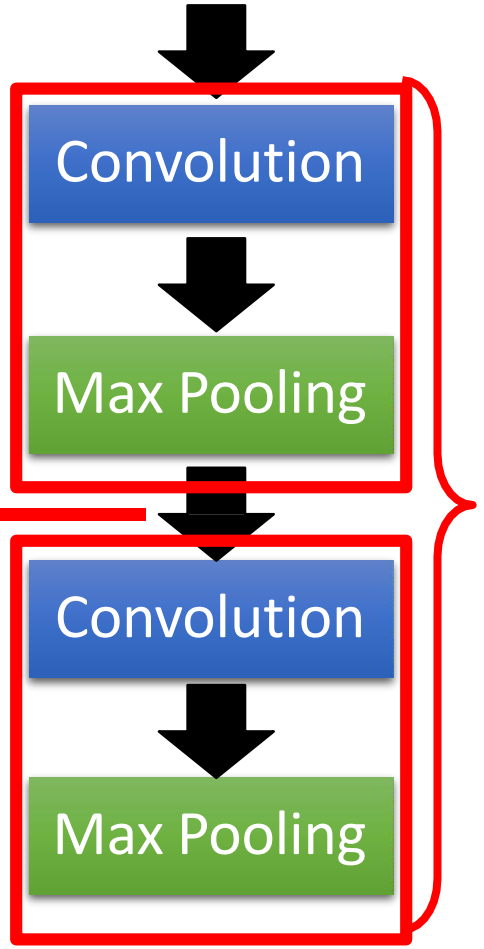
CNN网络结构


1 0 0 0 0 1
0 1 0 0 1 0
0 0 1 1 0 0
1 0 0 0 1 0
0 1 0 0 1 0
0 0 1 0 1 0
Conv
Max Pooling
New image but smaller
3 -1
0 1
3 0
1 3
6 x 6 image
2 x 2 image

CNN网络结构



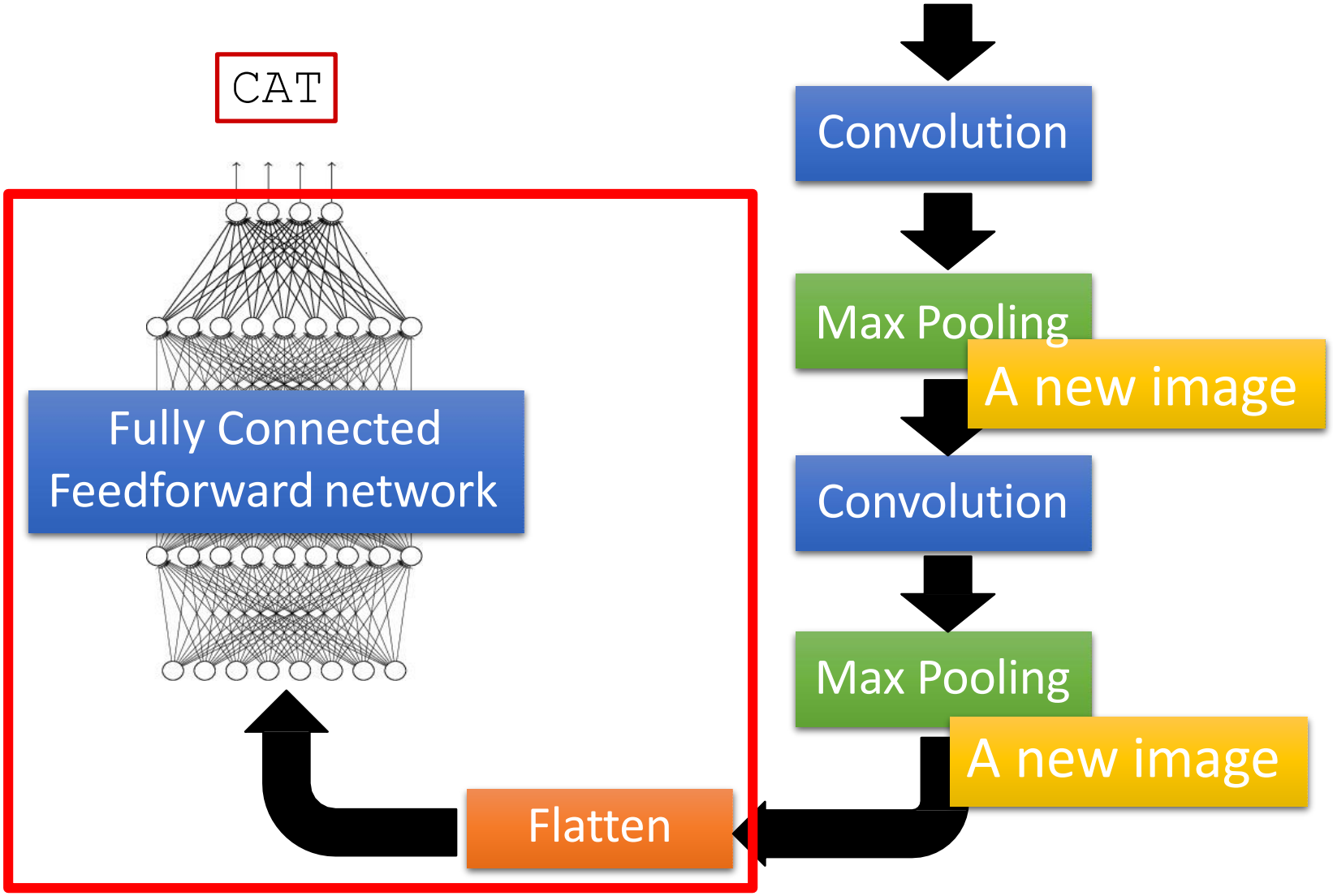




比原始图像更小

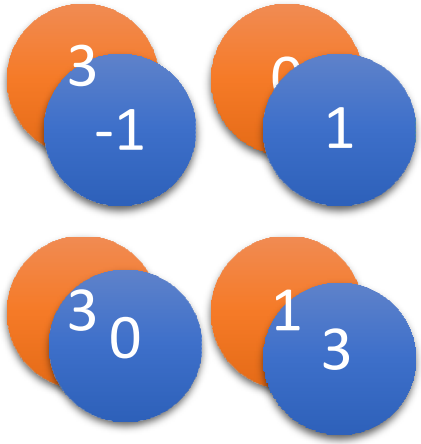
CNN网络结构




CAT
Convolution
Max Pooling
A new image
Convolution
Max Pooling
A new image
Flatten
Fully Connected
Feedforward network

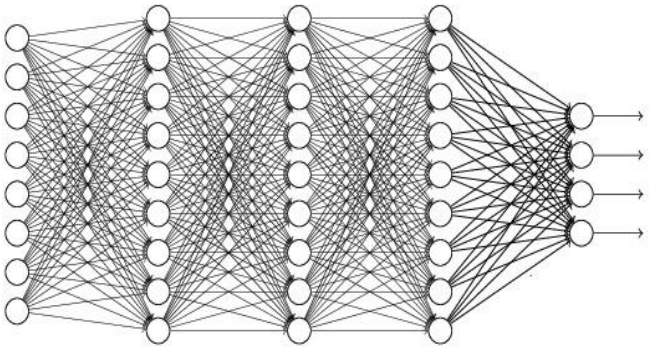
Flatten

- 特征图 -> 特征向量



Flatten





Fully Connected Feedforward network

循环神经网络（RNN）

神经网络的变体

卷积神经网络
(CNN)

循环神经网络（RNN)

具有记忆的神经网络

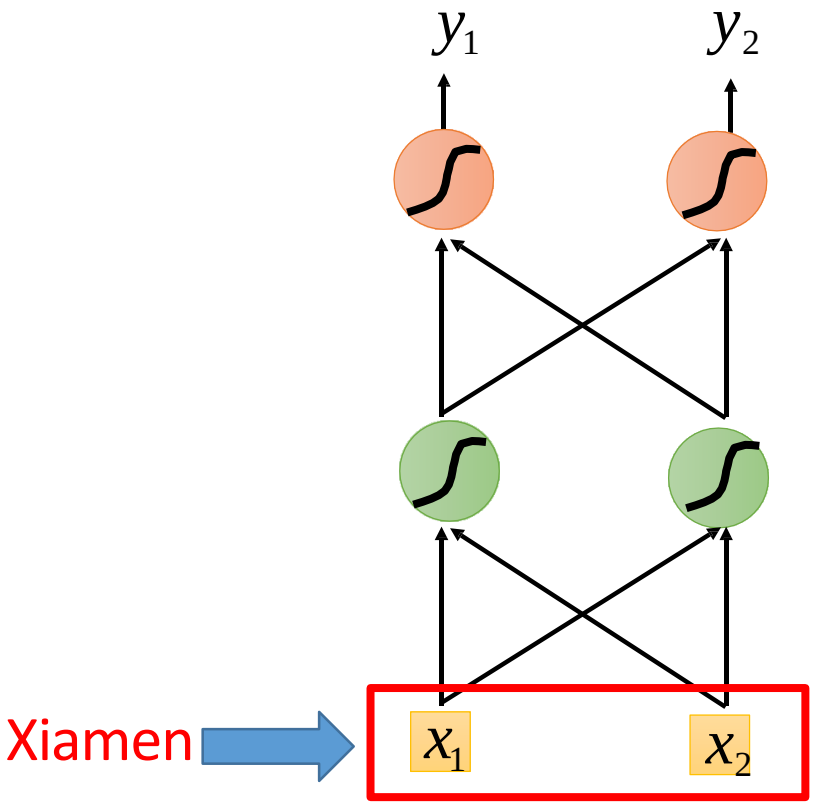
示例应用程序

- 填空问题


I would like to leave Xiamen on November 2nd.
智能订票系统
空位
地点:
Xiamen
时间:
November 2nd

示例应用程序

- 通过前馈神经网络解决填空问题?
  - 输入: a word (每个单词都被表示成一个向量)



1-of-N encoding

- 如何将每个单词表示成一个词向量?

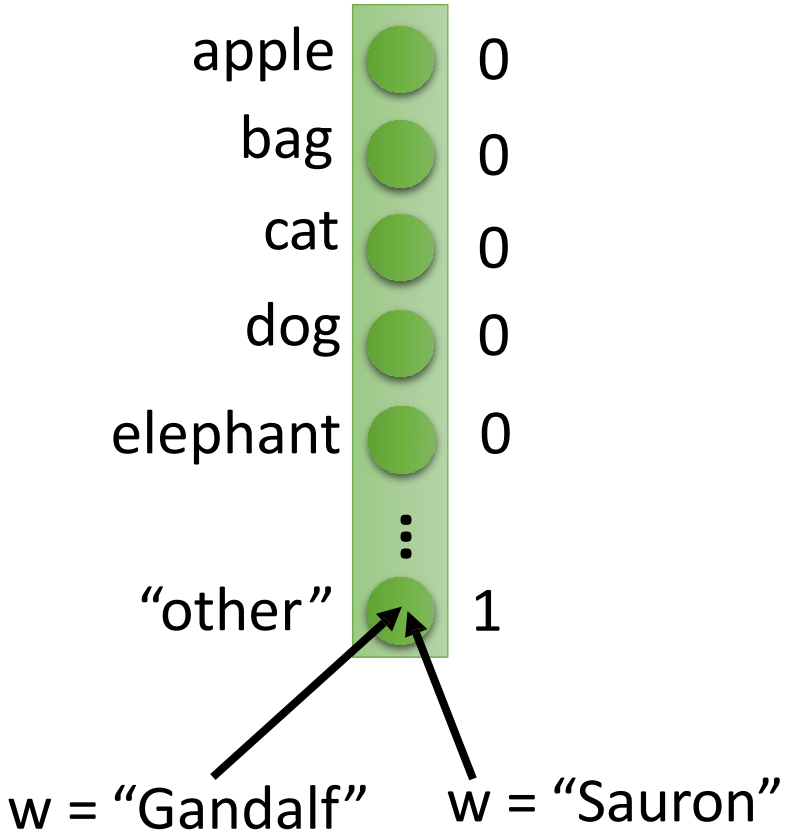
1-of-N Encoding 词典 = {apple, bag, cat, dog, elephant}

- 向量长度是词典大小
- 每个维度对应于词典中的一个单词
- 这个词的维度是1，其他是0

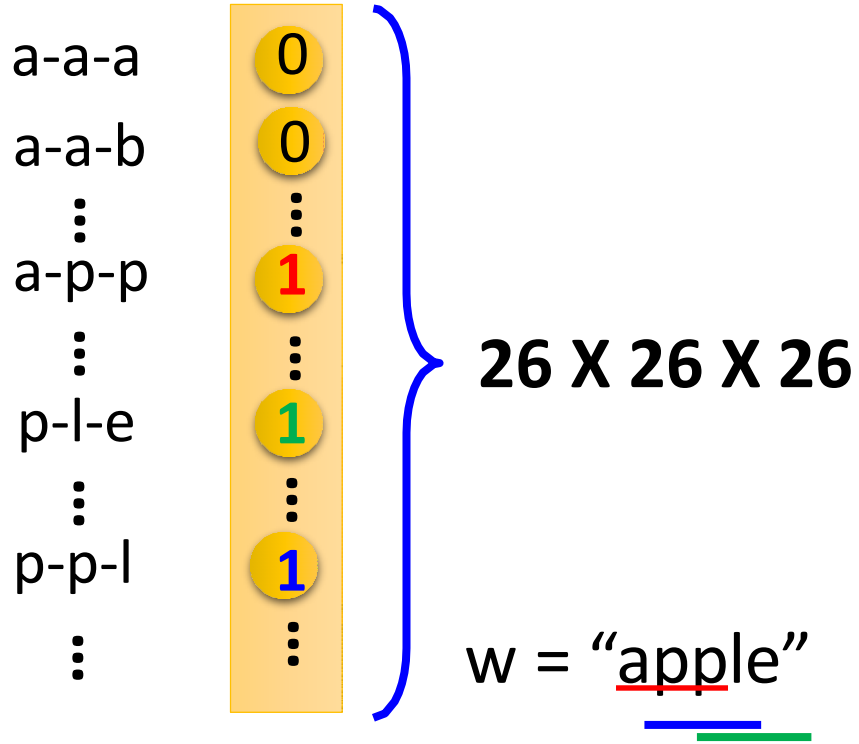
apple = [1 0 0 0 0]
bag = [0 1 0 0 0]
cat = [0 0 1 0 0]
dog = [0 0 0 1 0]
elephant = [0 0 0 0 1]

Beyond 1-of-N encoding

Dimension for "Other"

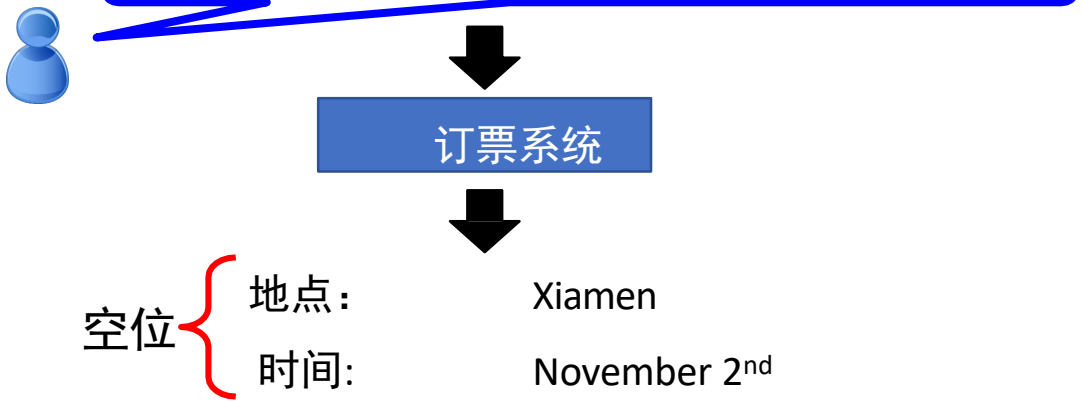


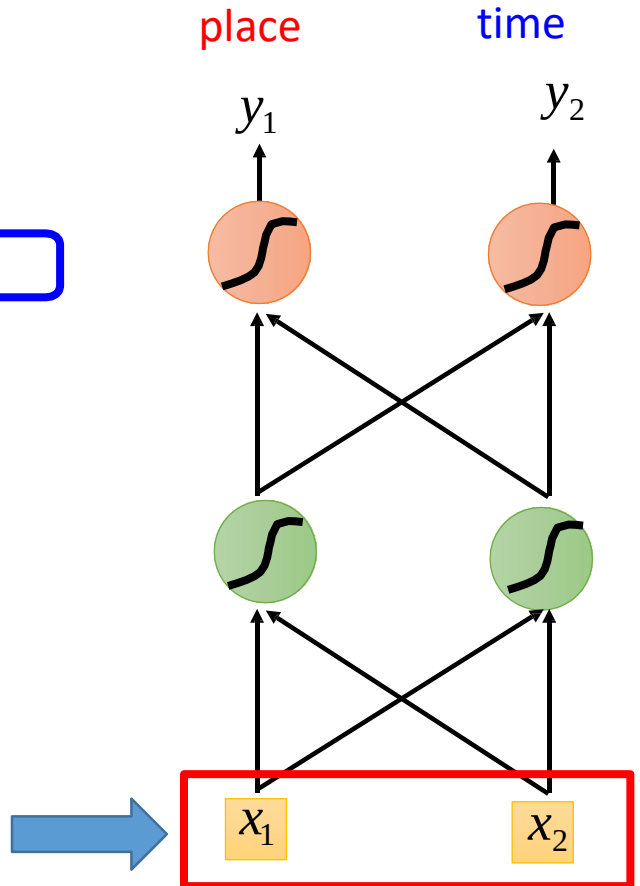
Word hashing



示例应用

- 通过前馈神经网络解决填空问题?
  - 输入：a word (每个单词都被表示成一个向量)
  - 输出：输入单词属于空位的概率分布





示例应用程序

leave Xiamen on November 2nd

other place other time time

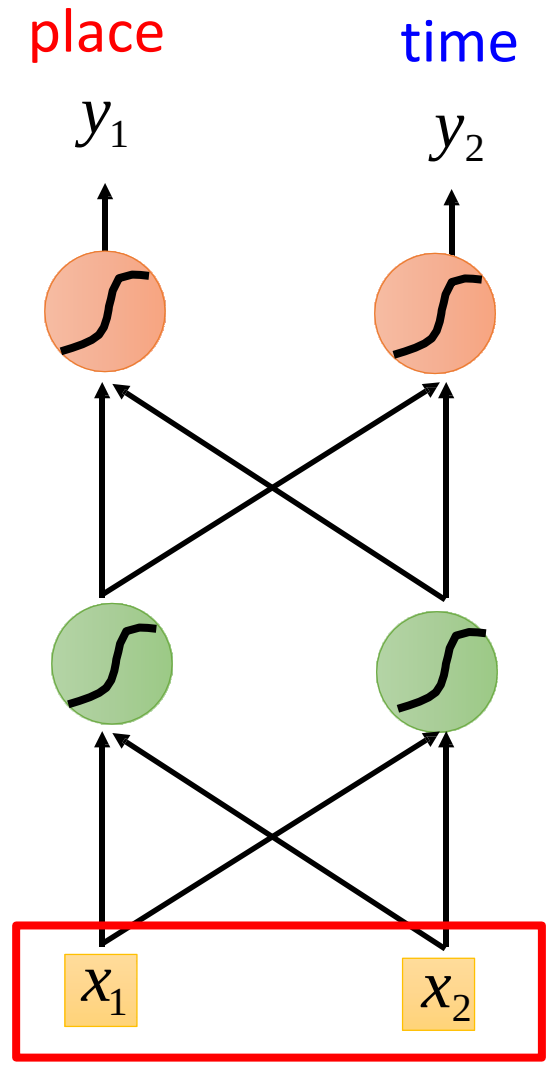
Problem?

leave Hangzhou on November 2nd

place

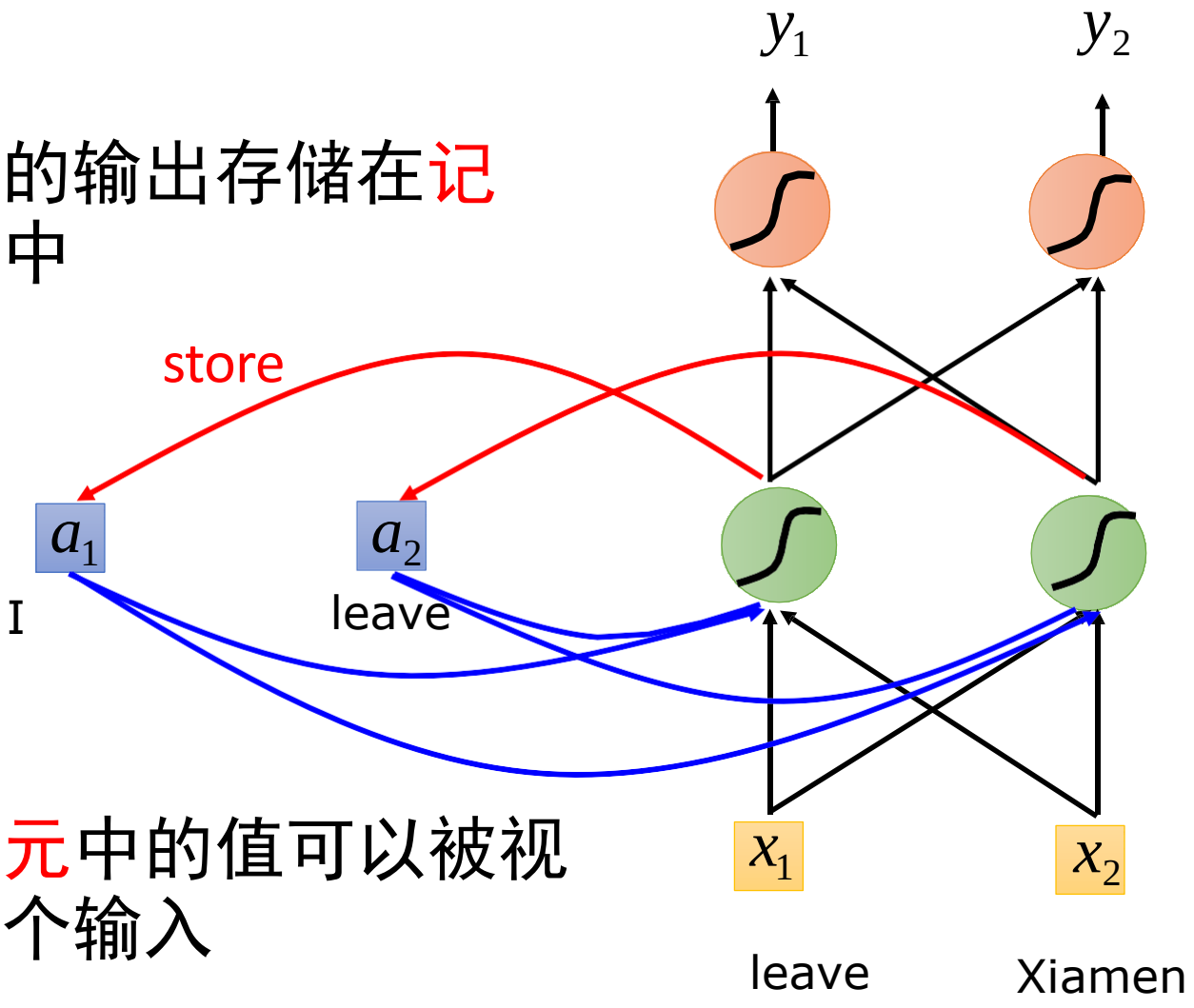
神经网络需要记忆

Xiamen



循环神经网络（RNN）

隐藏层的输出存储在记忆单元中



记忆单元中的值可以被视为另一个输入

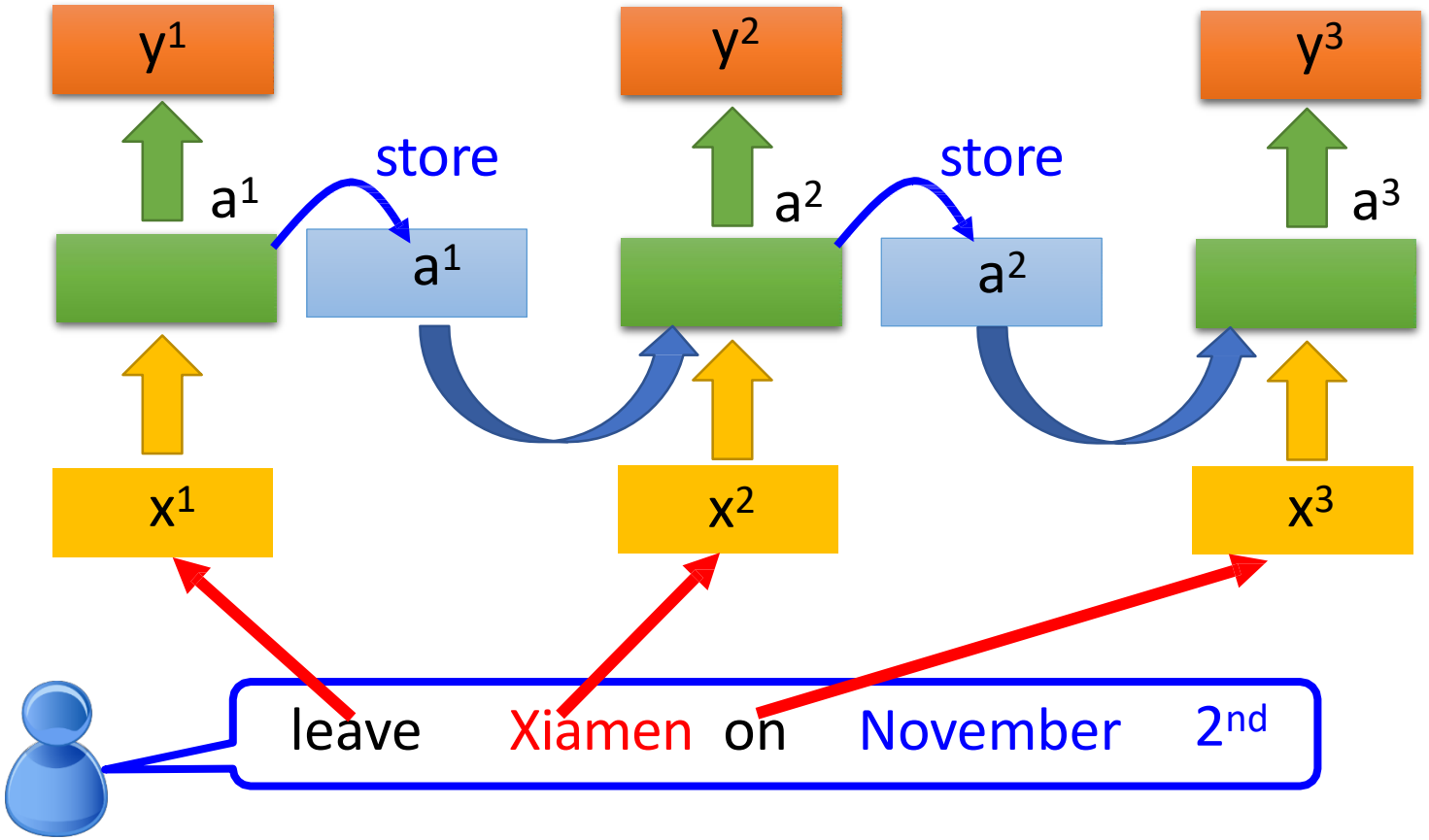
循环神经网络（RNN）

同样的网络多次使用

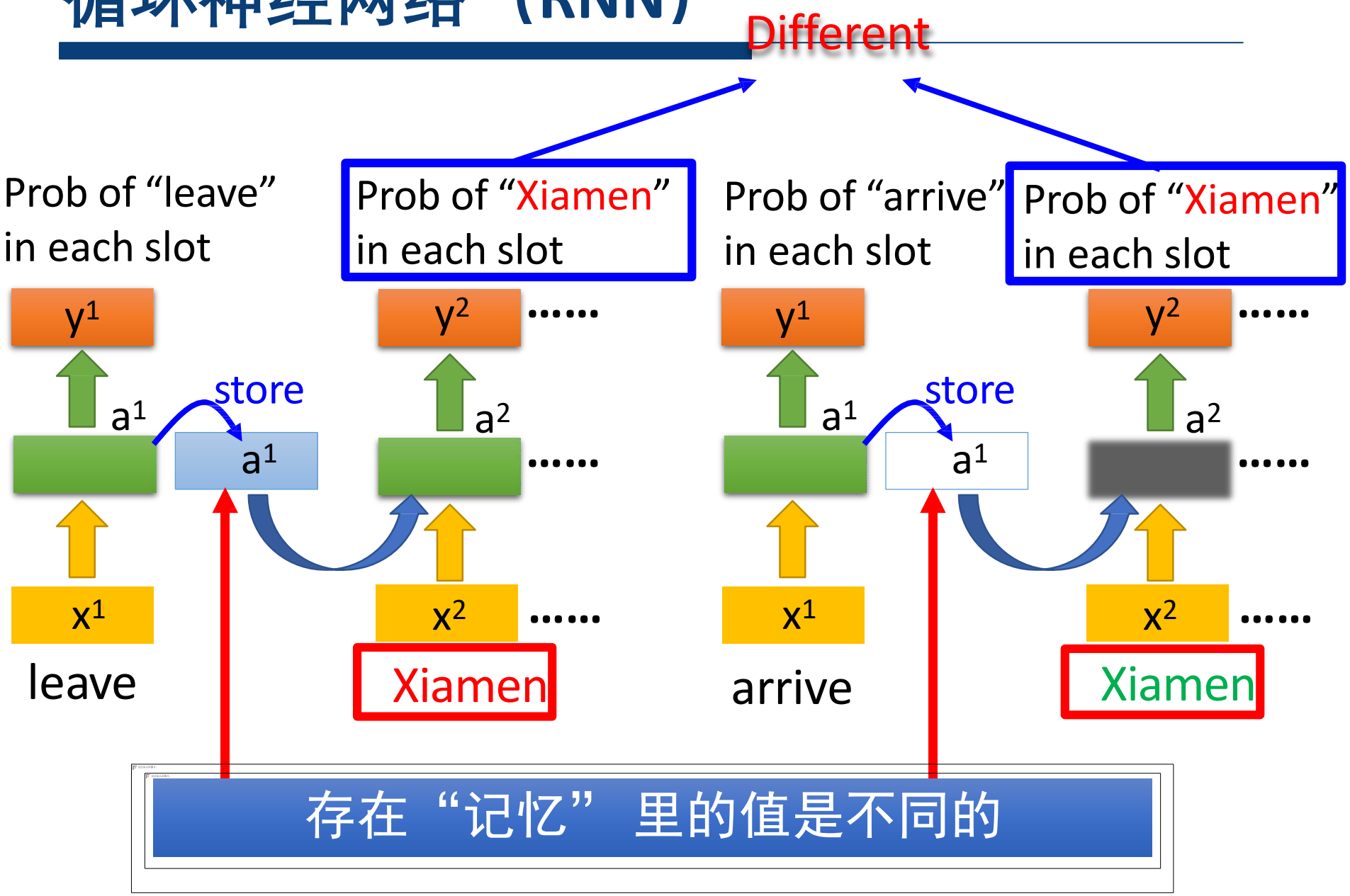
Probability of "arrive" in each slot

Probability of "Xiamen" in each slot

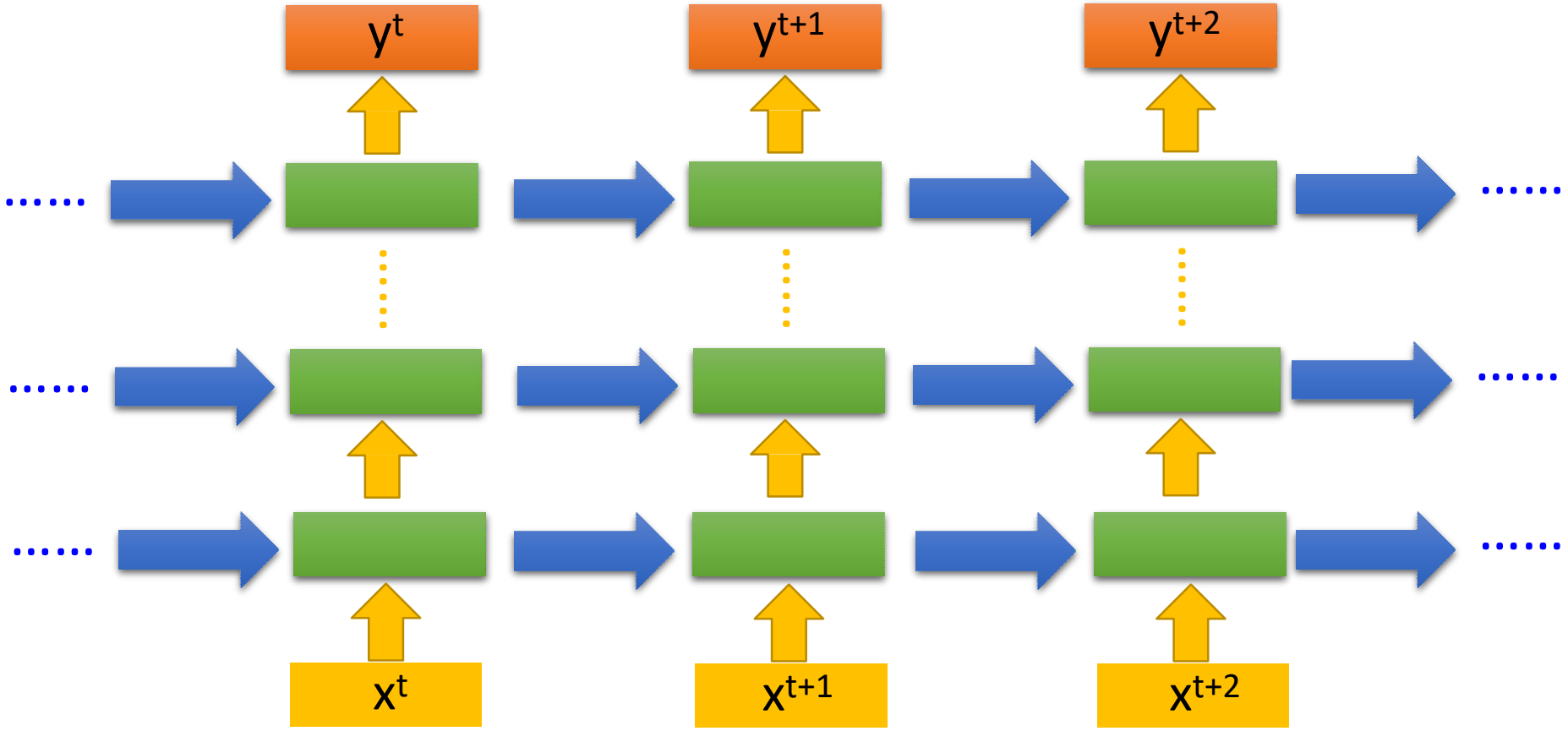
Probability of "on" in each slot



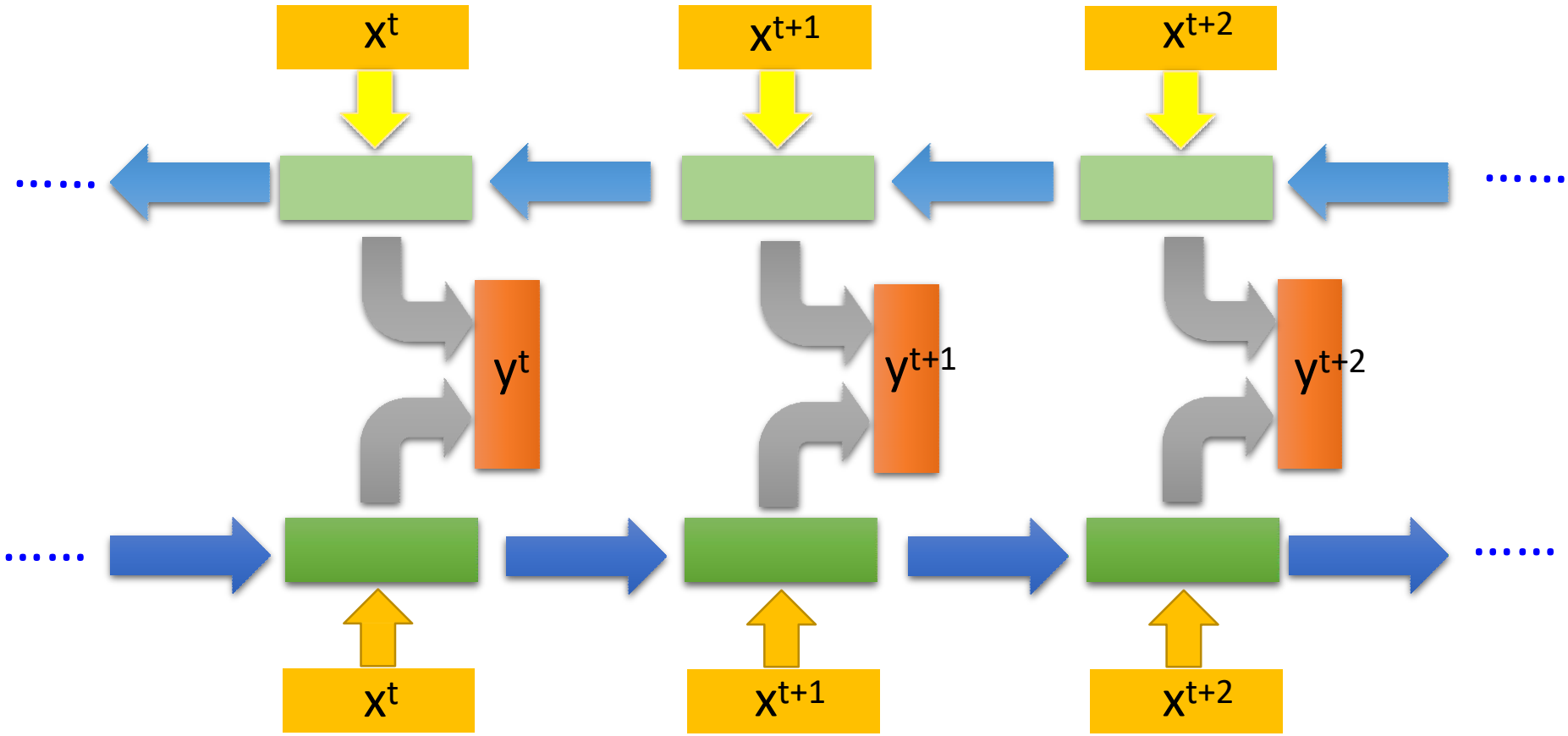
循环神经网络（RNN）


Different
Prob of “leave” in each slot
Prob of “Xiamen” in each slot
Prob of “arrive” in each slot
Prob of “Xiamen” in each slot
y1
y2
a1
a2
store
x1
x2
leave
Xiamen
arrive
Xiamen
存在“记忆”里的值是不同的

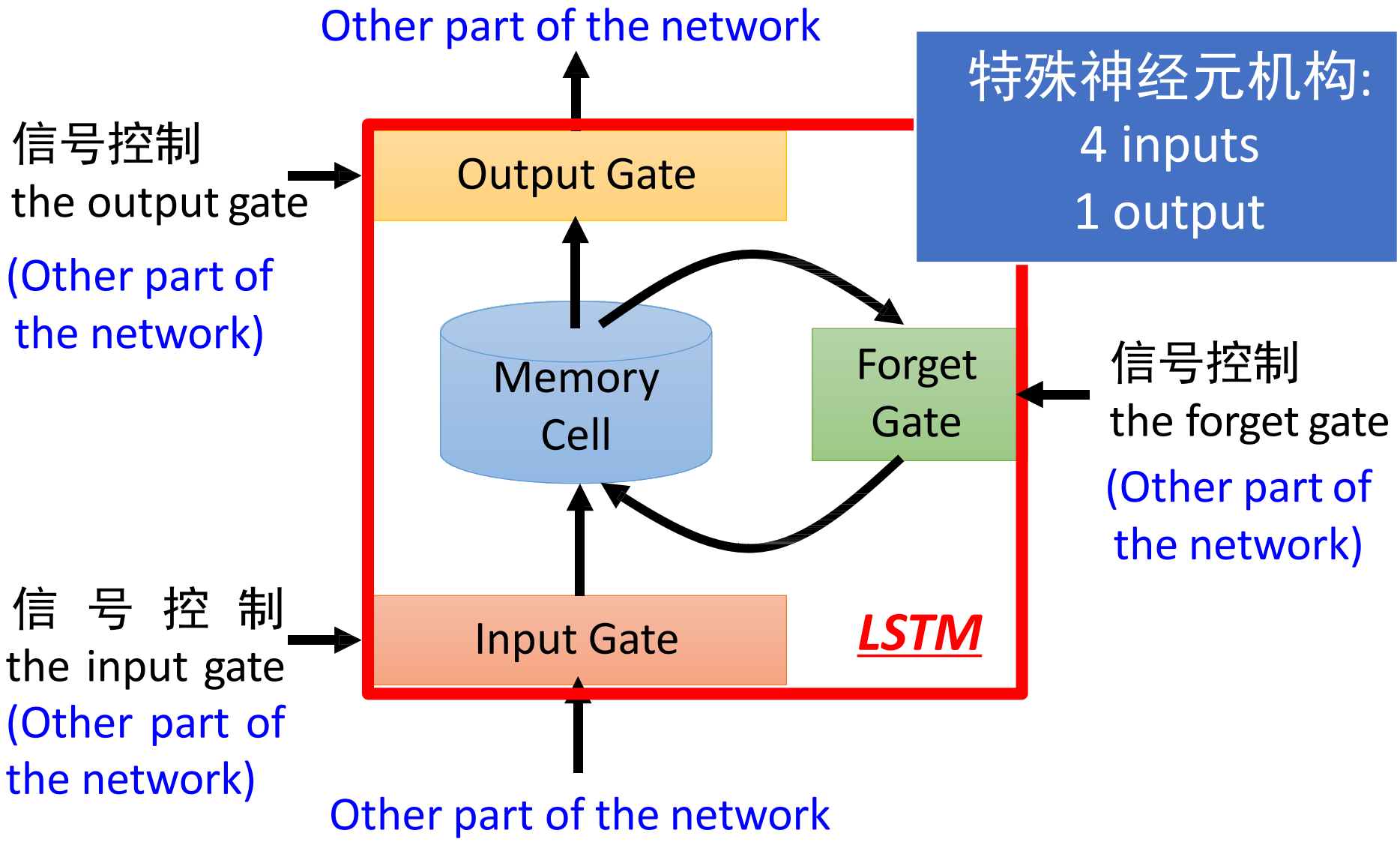
当然，它可以是深层网络 …



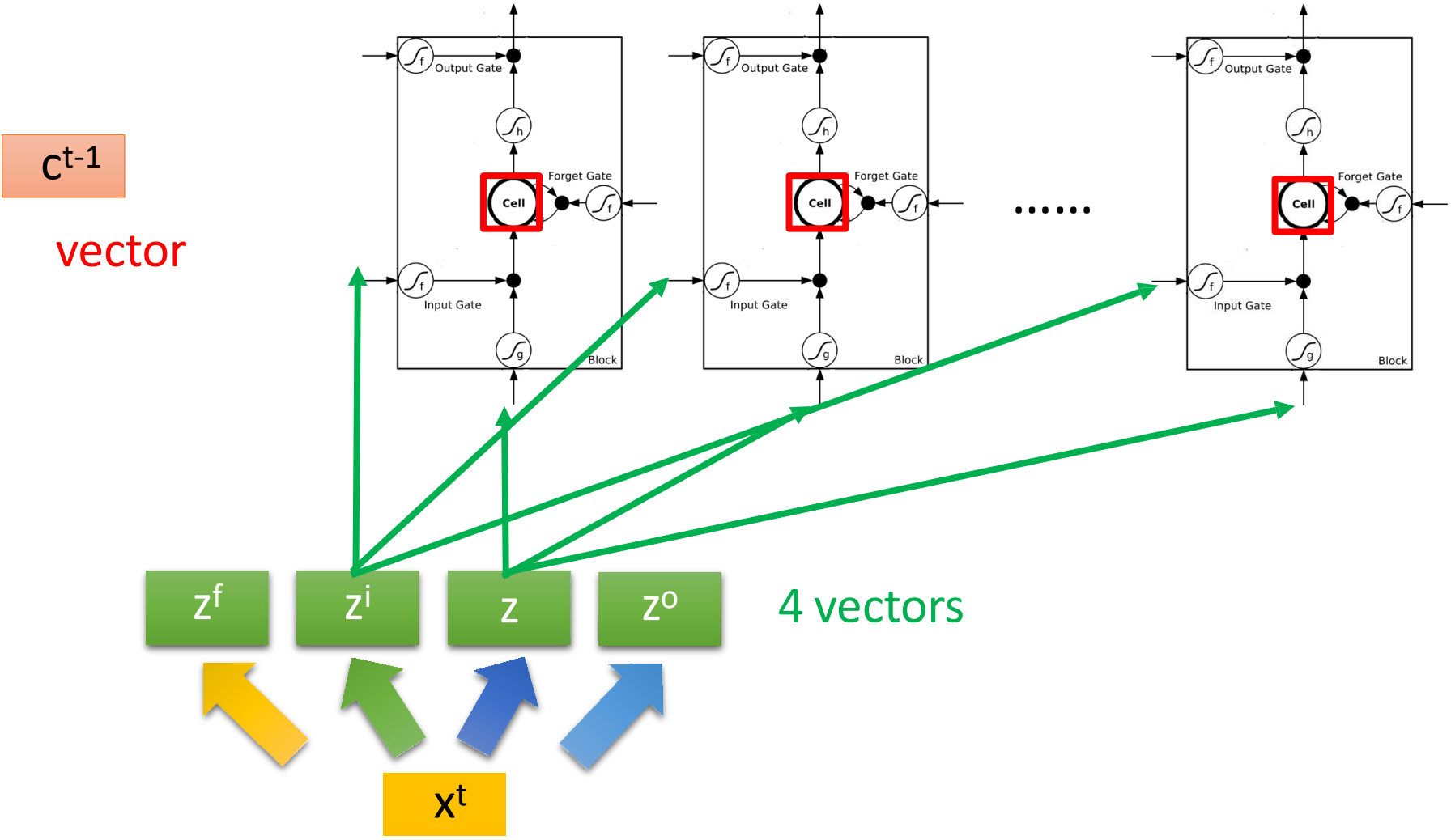
双向 RNN



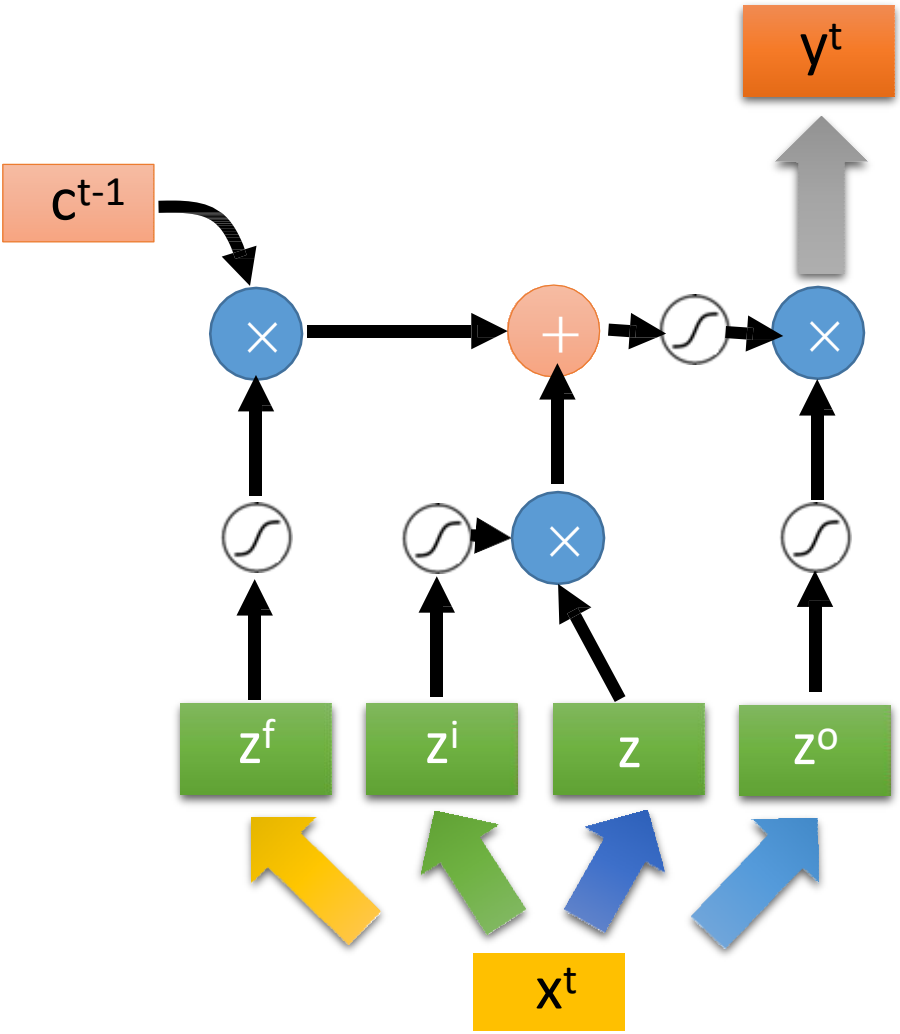
Long Short-term Memory (LSTM)

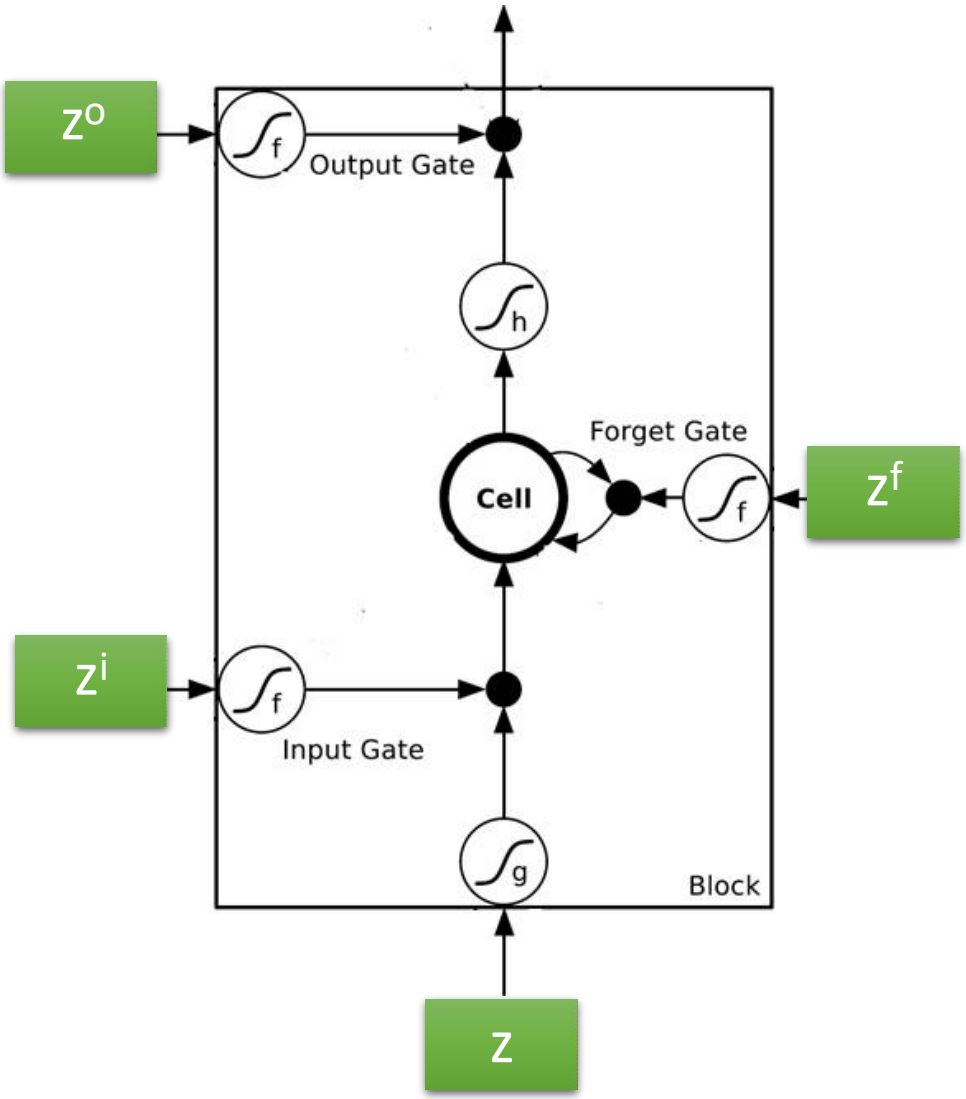


LSTM

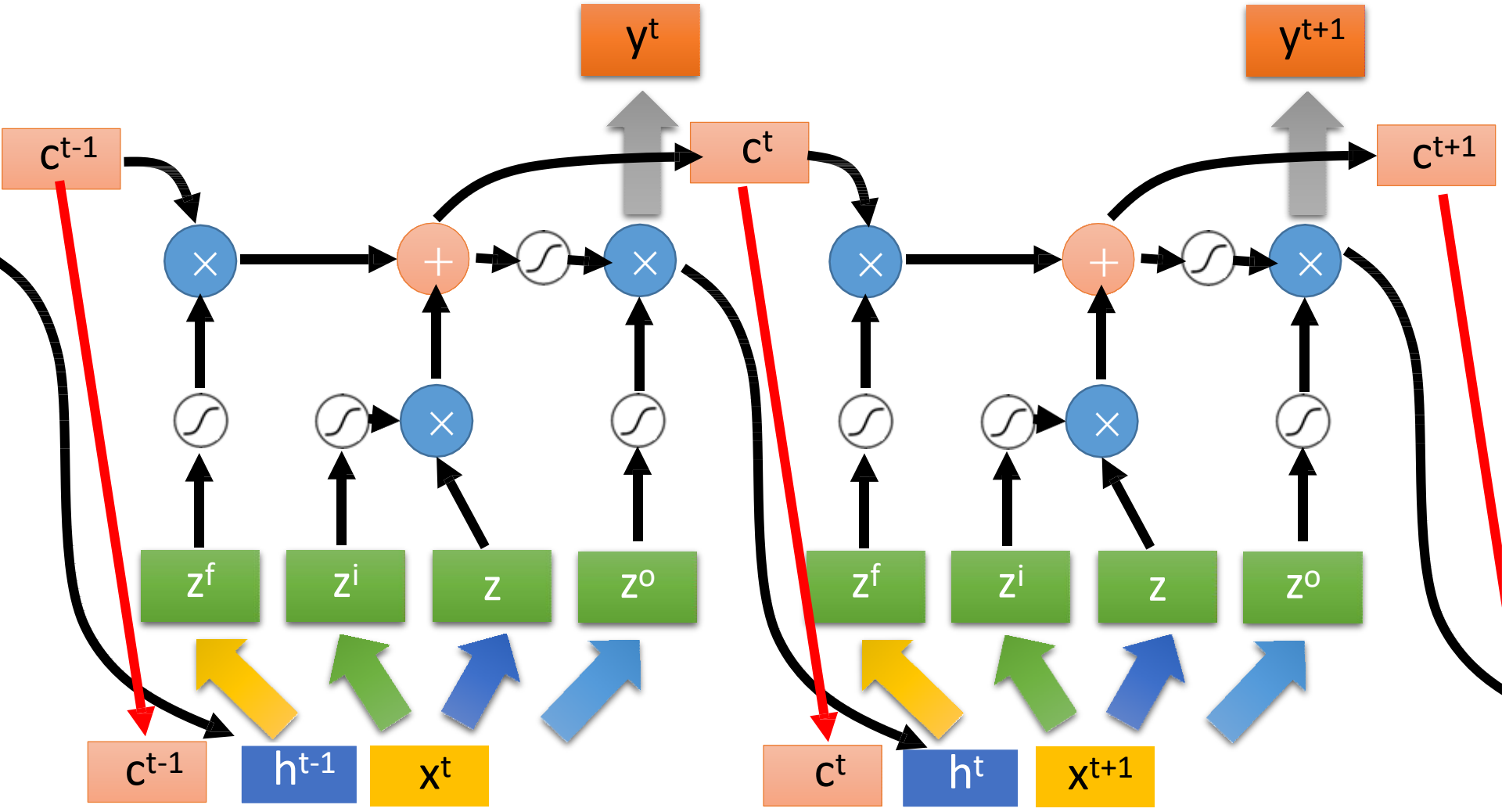

c^{t-1}
vector
Output Gate
Forget Gate
Cell
Input Gate
Block
.....
z^f
z^i
z
z^o
4 vectors
x^t

LSTM

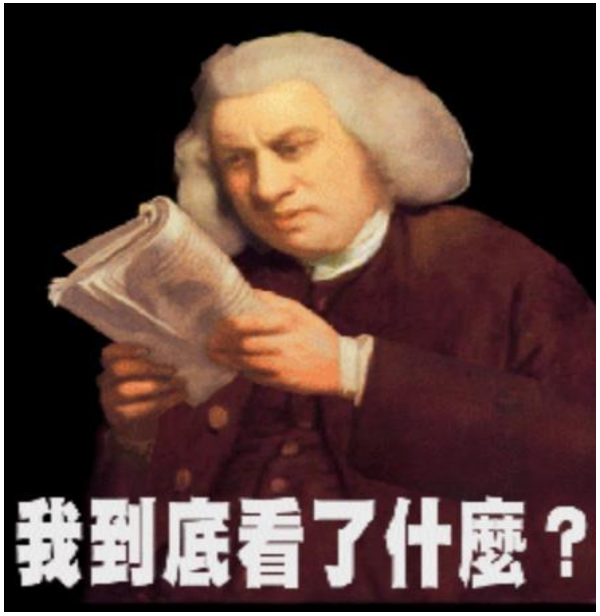

y^t
c^{t-1}
z^f
z^i
z
z^o
x^t

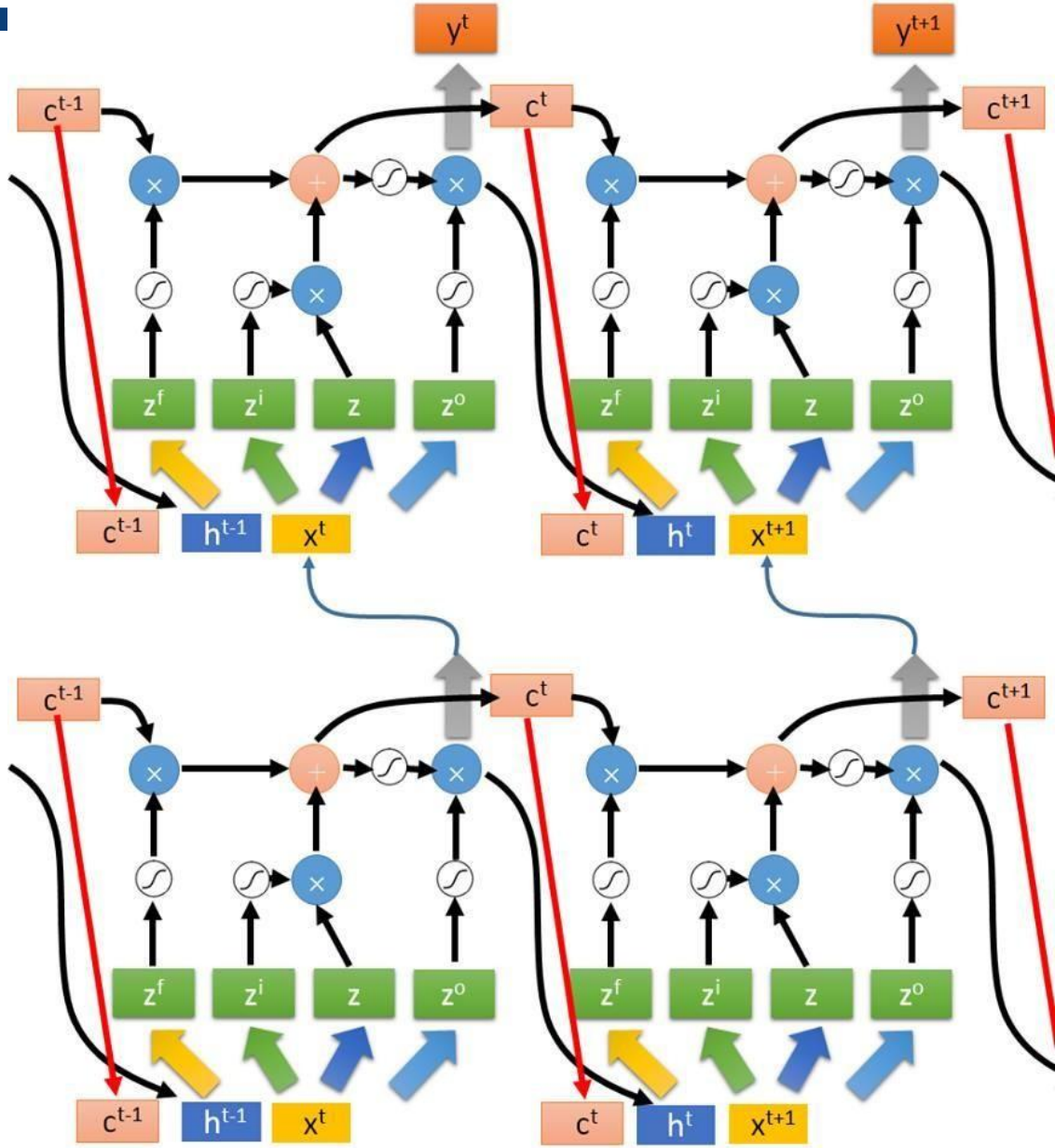

z^o
Output Gate
h
Forget Gate
Cell
z^f
z^i
Input Gate
g
Block
z

LSTM

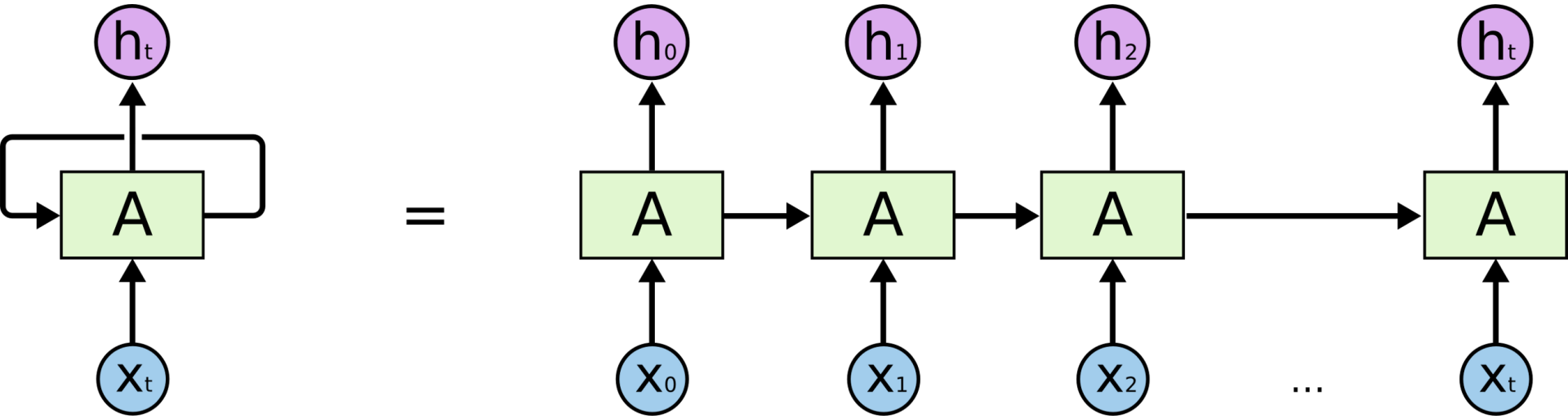


Multiple-layer LSTM

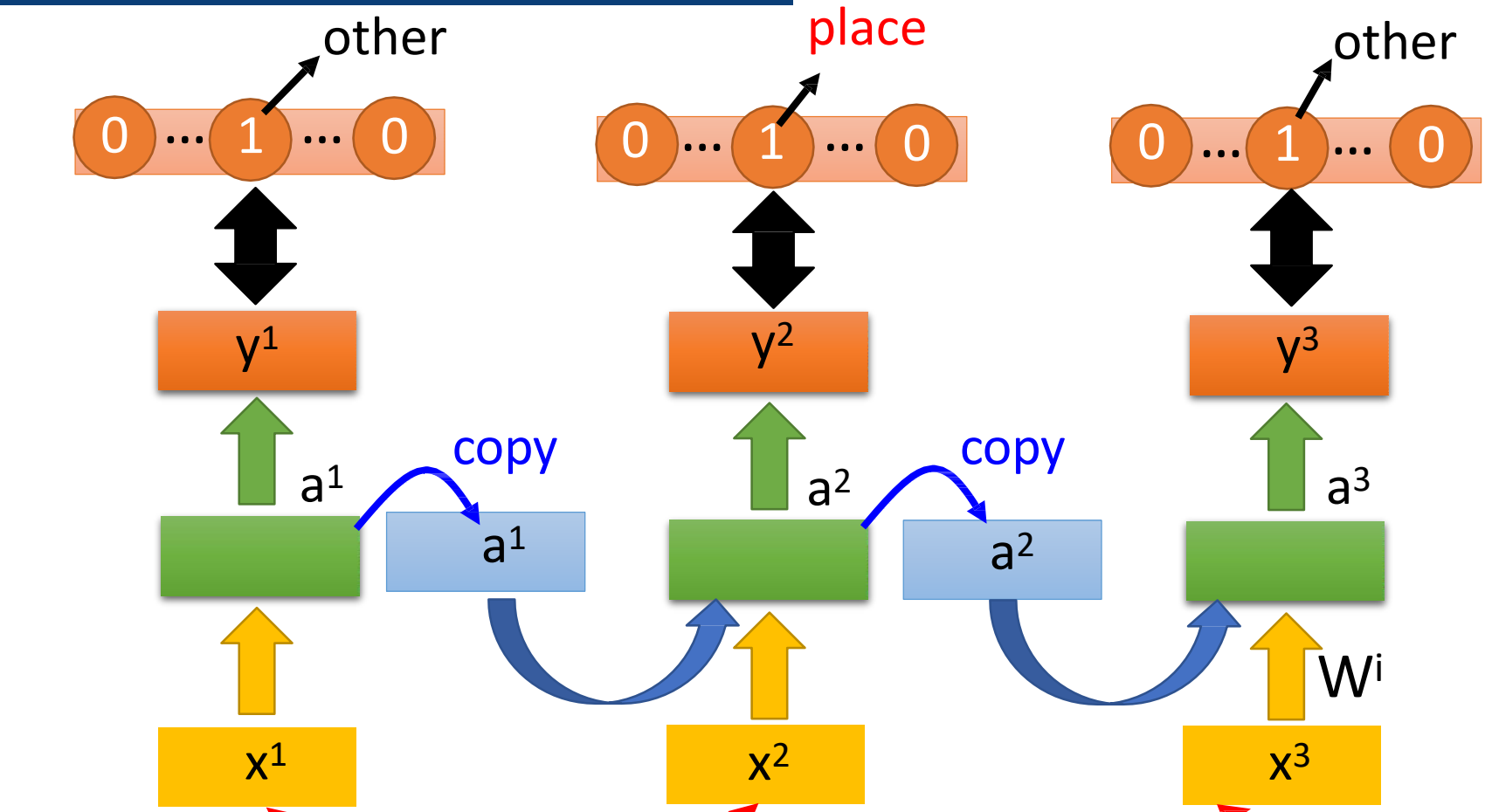




RNN结构

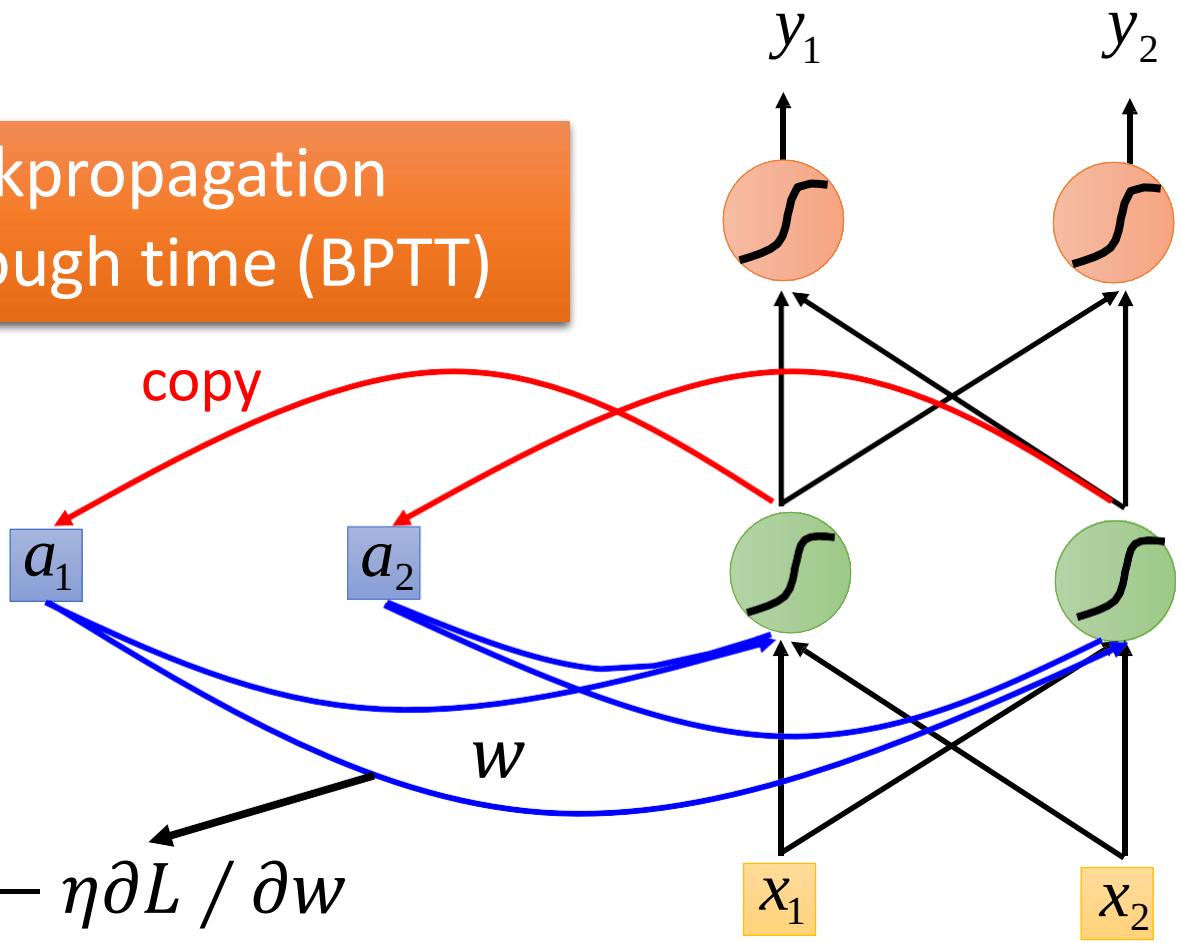


Learning Target



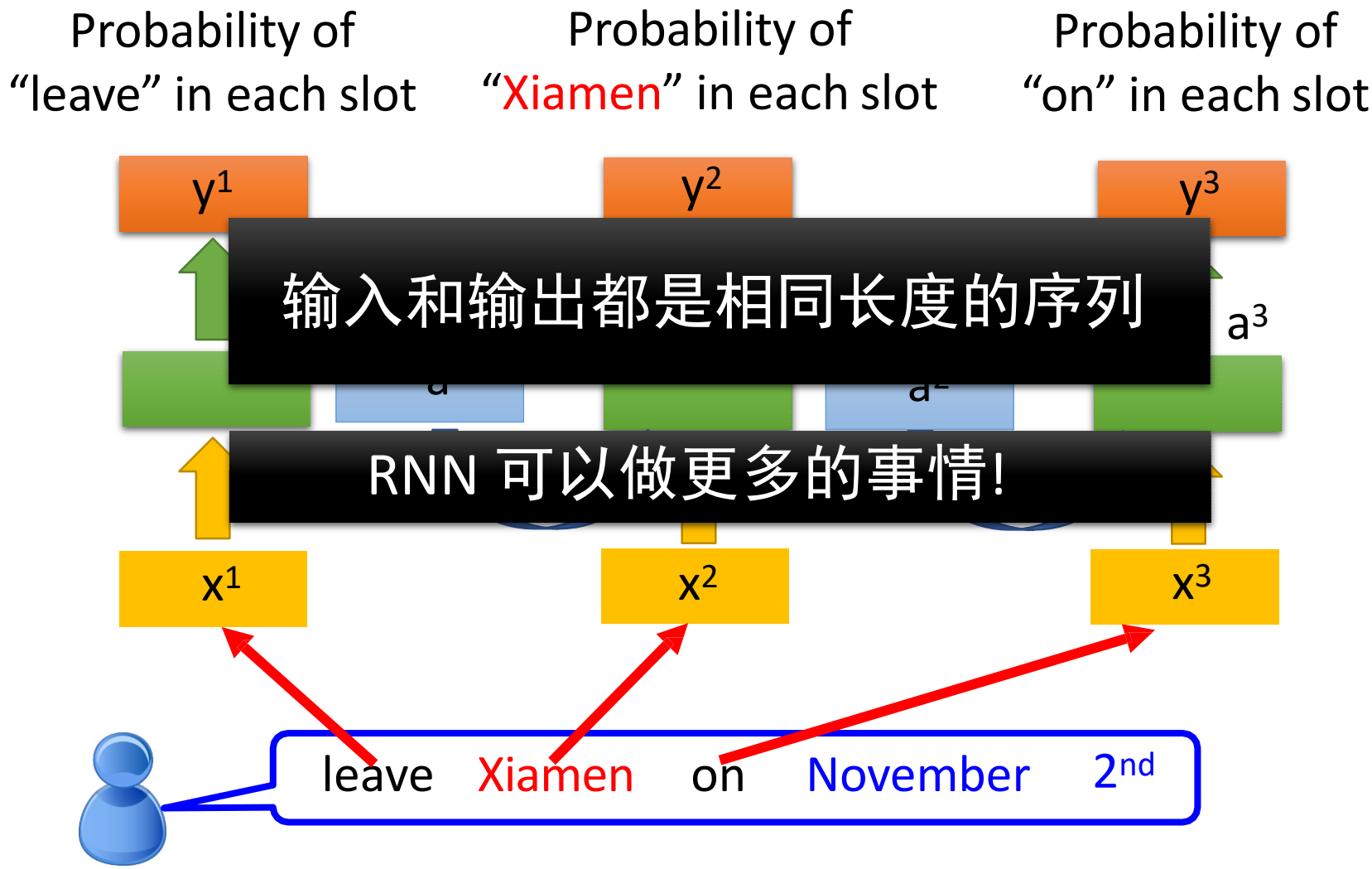
Learning

Backpropagation through time (BPTT)



RNN的学习在实践中是非常困难的

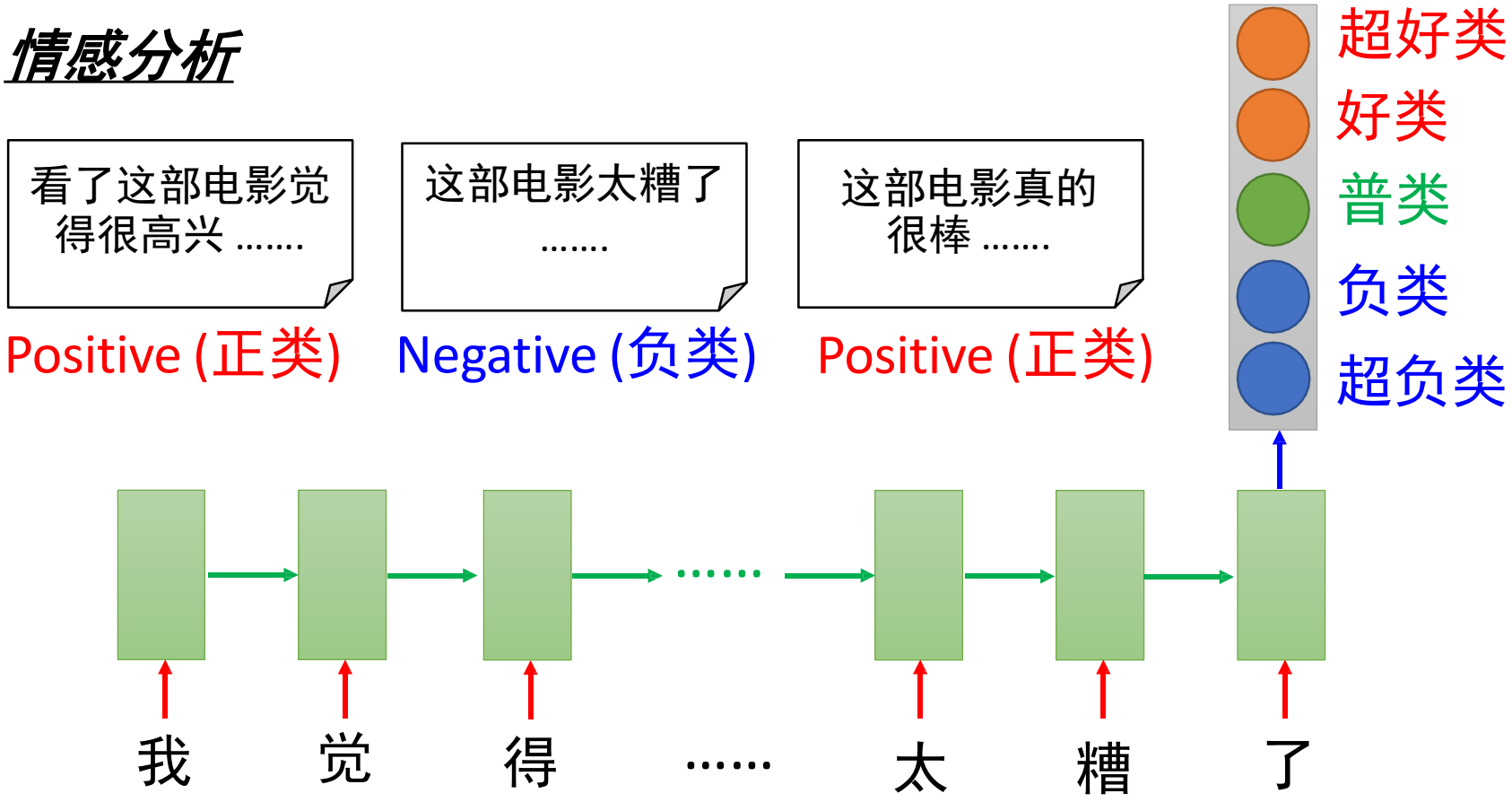
更多应用


Probability of "leave" in each slot
Probability of "Xiamen" in each slot
Probability of "on" in each slot
y1
y2
y3
a3
输入和输出都是相同长度的序列
RNN 可以做更多的事情!
x1
x2
x3
leave
Xiamen
on
November
2nd

Many to one

- 输入是一个向量序列，但输出只是一个向量

情感分析



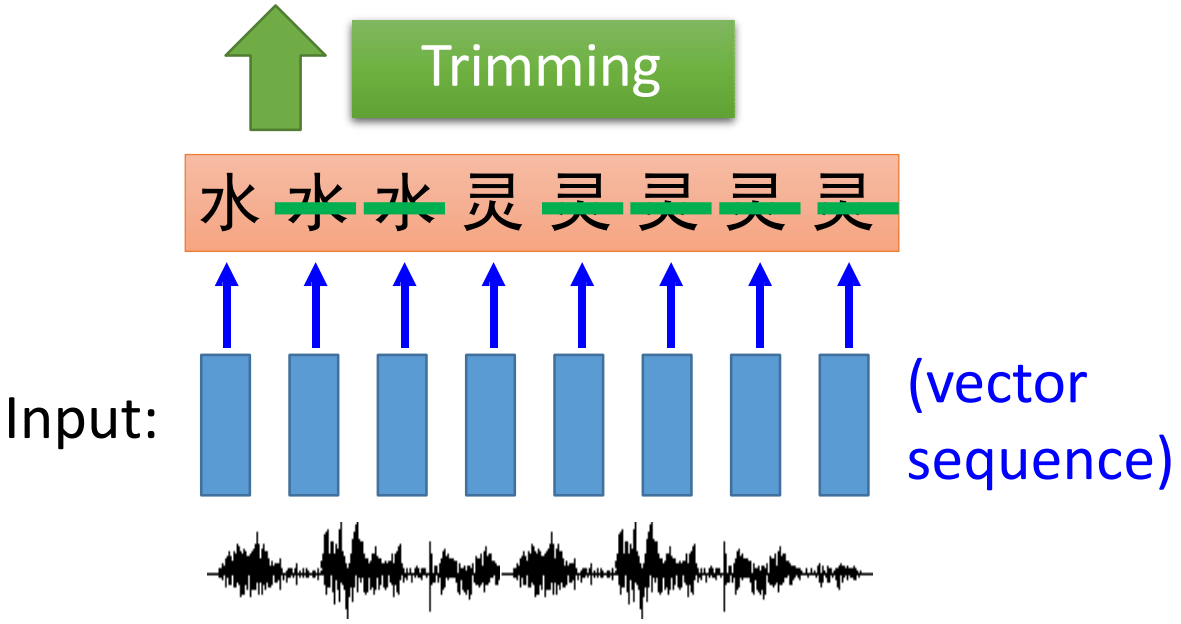
Many to Many (输出较短)

- 输入和输出都是序列, ***但输出比较短***
  - E.g. 语音识别

Output: “水灵” (character sequence)

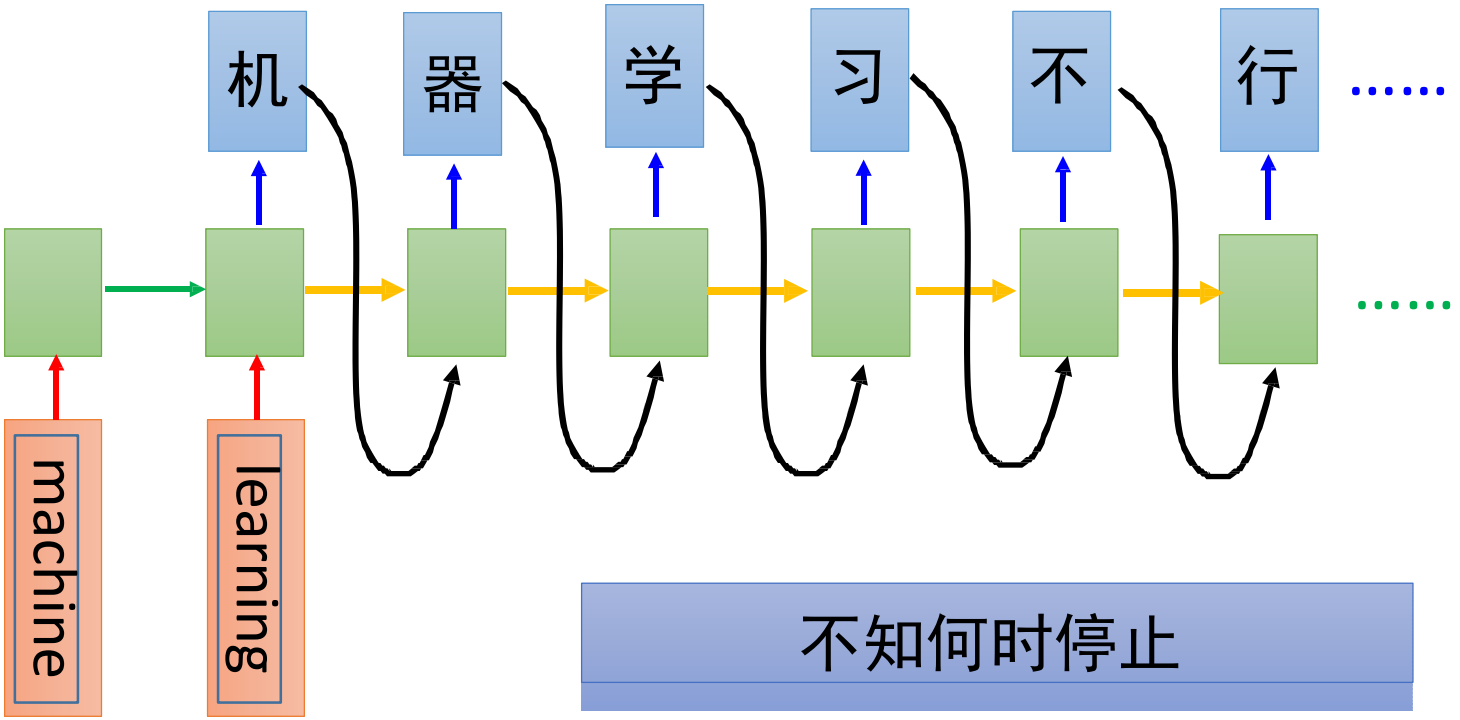
Problem?

Why can’t it be “水灵灵”



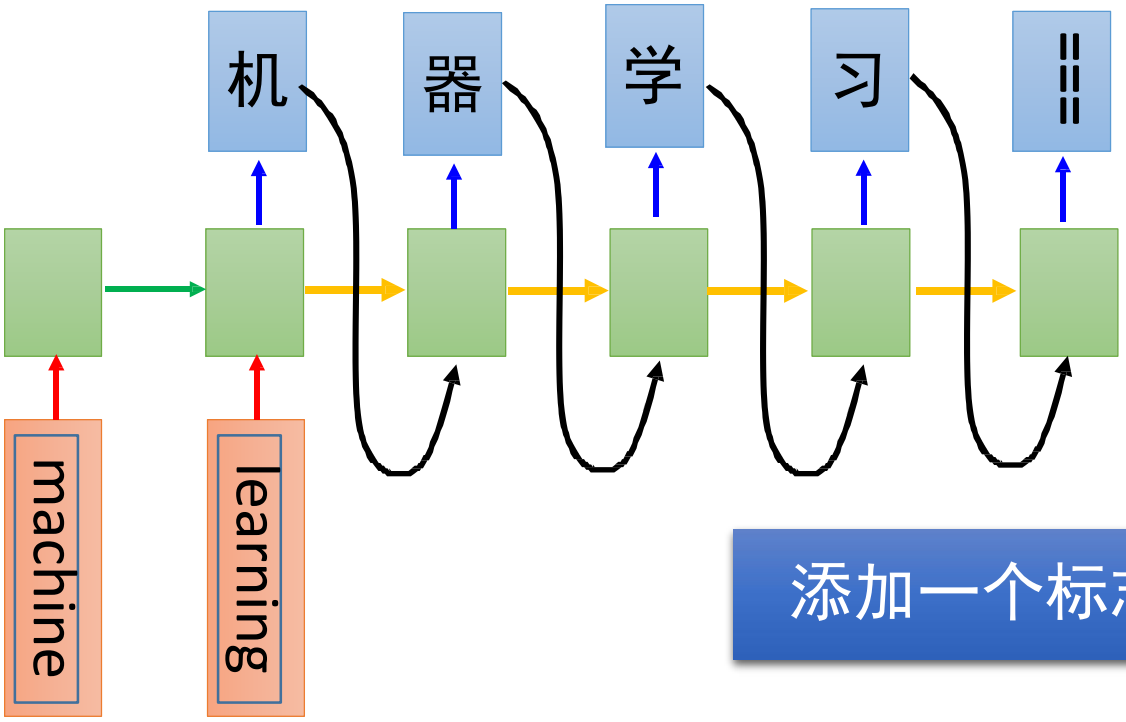
Many to Many (No Limitation)

- 输入和输出都是不同长度的序列 → ***Sequence to sequence learning***
  - E.g. ***机器翻译*** (machine learning→机器学习)



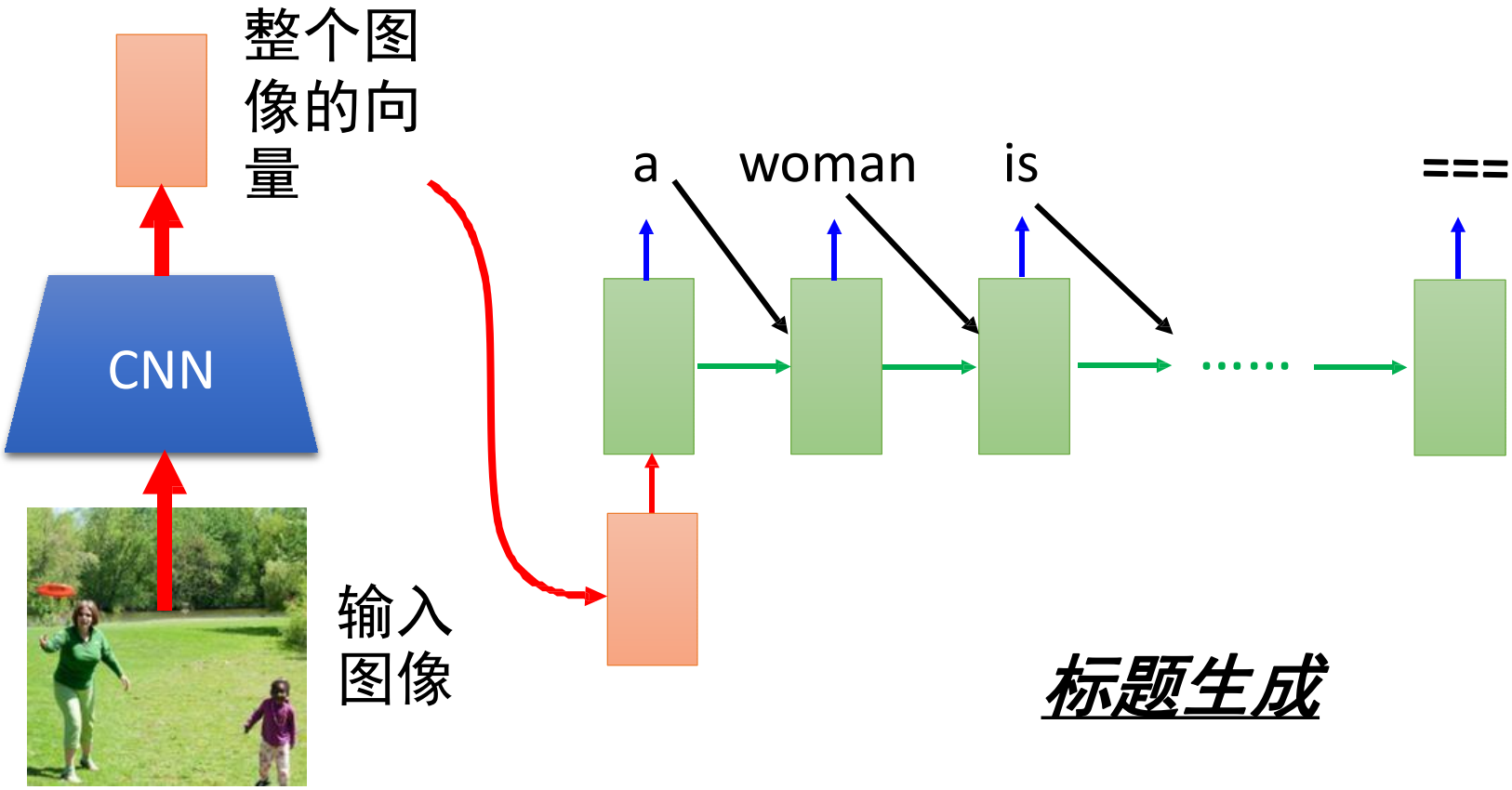
Many to Many (没有限制)

- 输入和输出都是不同长度的序列 → ***Sequence to sequence learning***
  - E.g. ***机器翻译*** (machine learning→机器学习)



One to Many

- 输入图像，但输出一系列单词



Demo

- **Image Captioning**
  - **https://image-to-caption.com**
  - **Image Caption Generator (Free Online Tool)**
    - Prof. Cheng WANG
    - 皮卡丘
    - 皮卡
    - Selfie Lisa !

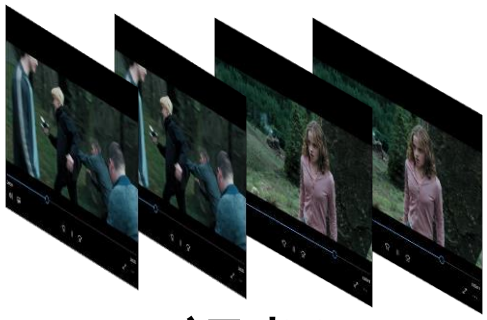
Microsoft

CaptionBot

I am not really confident, but I think it's Mona Lisa et al. taking a selfie.



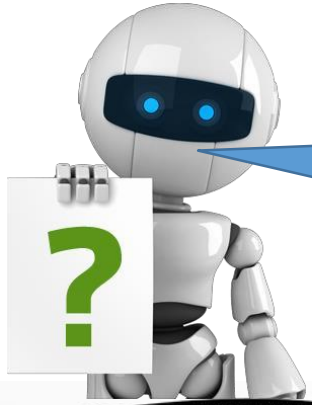
应用：视频字幕生成



视频



一群人被一棵树撞到



一个女孩子正在奔跑





一群人走在森林里

Demo

- Sketch-RNN
  - https://magenta.tensorflow.org/assets/sketch_rnn_demo/index.html
  - Cat and Fish

End

